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Effects of fluid shear-thinning and yield stress on free convection in concentric and eccentric cylindrical annuli

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ABSTRACT

Laminar free convection in yield-pseudoplastic fluids (the Herschel-Bulkley model) for concentric and eccentric cylindrical annuli has been studied numerically. The combined effects of shear-thinning viscosity and yield stress on the heat transfer characteristics have been examined for the following ranges of parameters: Rayleigh number, Ra (10^3 to 10^6), Oldroyd number, Od (0 to Od_{max}), power-law index, n (0.2–1), Prandtl number, Pr (10–100), eccentricity, ε (0–1.18) and angular position of the inner cylinder, ϕ (0° to 180°). The results are interpreted in terms of the yield surfaces, fraction of yielded fluid, streamlines, isotherms, local Nusselt number and average Nusselt number. Overall, the eccentric positioning of the heated cylinder along the vertical centreline fosters convective transport with reference to that for the case of the concentric annulus. It is possible to achieve augmentation of up to 30% in heat transfer for the case of $\varepsilon = 1.18$, $\phi = 0°$ with respect to the concentric case at $Ra = 10^5$, $Pr = 10$. This is ascribed to the greater fraction of the annular area occupied by the yielded fluid-like regions. Conversely, horizontal shifting of the inner cylinder to an off-centre position has an adverse effect on heat transfer. Limited transient simulations have also been run to identify the conditions for the loss of steady flow behaviour. A predictive correlation has been developed for the average Nusselt number for its estimation in new applications.

1. Introduction

Free convection heat transfer in a fluid medium confined in an annulus has been one of the extensively studied configurations owing to its fundamental significance and wide-ranging practical applications in heat transfer equipment, thermal energy storage devices, cooling of electronic units, appliances, cables and transmission lines, etc. [1–3]. Additional examples are found in the context of horizontal oil drilling operations wherein the drill pipe often tends to sag thereby leading to eccentricity of the annulus [4]. Similarly, processed foods like carrots, radishes, etc. are reheated in cylindrical cans and retorts and with multiple rods it can naturally lead to eccentricity in the annulus. Further potential applications are encountered in the use of nanofluids and/or smart lubricants, etc. in the cooling of electronic components, nuclear reactor fuel rod, and in the processing of pulp and fibre suspensions in various heat exchange equipment. In addition, eccentric annulus can also lead to the lowering of pressure drop and/or enhancement in heat transfer [5]. Finally, eccentricity can also occur due to the thermal expansion and manufacturing tolerances.

In the thermal processing of foodstuffs such as in sugar making, the thickening of soups and other food products, indirect heating is often used by submerging a bundle of cylinders carrying steam in the mother liquor. Naturally, each cylinder of the bundle cannot form an eccentric annulus. In the context of yield stress fluids, where heat transfer tends to be intrinsically slow, the asymmetric configurations have also been shown to alleviate this difficulty to a certain extent, e.g., see Ref. [6]. Asymmetry of the flow field induced by employing off-centre or side-entering impellers also leads to enhanced mixing of such fluids [7]. Thus, the present work has been motivated by such pragmatic and theoretical considerations, especially with regard to the role of eccentricity in the context of yield stress fluids. Much literature exists covering various facets of this configuration such as the influence of eccentricity, aspect ratio (ratio of the two radii), wall boundary conditions (isothermal, isoflux and mixed), flow transition characteristics, etc., on the free convective momentum and heat transfer characteristics of Newtonian fluids. Since the pertinent studies have been reviewed extensively elsewhere [1,8–11], only the prominent investigations are

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Nomenclature

A_s	Surface area of the inner cylinder, m^2
C_P	Specific heat capacity, $J\ kg^{-1}\ K^{-1}$
C_D	Drag coefficient, *dimensionless*
E	Centre to centre distance between the two cylinders, m
F_D	Total drag force per unit length of the cylinder, N/m
g	Acceleration due to gravity, ms^{-2}
Gr	Grashof number, *dimensionless*
Gr^*	Modified Grashof number, *dimensionless*
k	Thermal conductivity of fluid, $W\ m^{-1}\ K^{-1}$
k_{eq}	Mean equivalent conductivity, *dimensionless*
L	Arc length, dimensionless
M_P	Papanastastiou regularization parameter, *dimensionless*
m	Fluid consistency index, $Pa\ s^n$
n	Power-law index, *dimensionless*
N_E	Total number of elements in the computational domain, *dimensionless*
N_S	Number of elements on the surface of the inner cylinder, *dimensionless*
Nu_θ	Local Nusselt number, *dimensionless*
Nu_{avg}	Average Nusselt number, *dimensionless*
Nu_∞	Average Nusselt number in the conduction limit, *dimensionless*
Od	Oldroyd number, *dimensionless*
p	Pressure, *dimensionless*
Pr	Prandtl number, *dimensionless*
Pr^*	Modified Prandtl number, *dimensionless*
Ra	Rayleigh number, *dimensionless*
R_i	Radius of the inner cylinder, m
R_o	Radius of the outer cylinder, m
T_C	Temperature of the outer cylinder, K
T_H	Temperature of the inner cylinder, K
$\boldsymbol{V}$	Velocity vector, *dimensionless*
V_C	Characteristic velocity for buoyancy-induced flow, $m\ s^{-1}$
Y	Yield number, *dimensionless*
x, y	Cartesian coordinates, m
u, v	x- and y-components of the velocity, ms^{-1}

Greek symbols

α	Thermal diffusivity of fluid, $m^2\ s^{-1}$
β	Coefficient of thermal expansion, K^{-1}
$\dot{\gamma}$	Rate of strain tensor, *dimensionless*
ε	Eccentricity of the annulus, *dimensionless*
η	Apparent viscosity of the fluid, *dimensionless*
Θ	Non-dimensional temperature, *dimensionless*
λ_{BE}	Regularization parameter in Bercovier-Engelman model, *dimensionless*
μ_{HB}	Viscoplastic fluid viscosity, *dimensionless*
μ_y	Yielding viscosity, *dimensionless*
ξ	Fraction of yielded zone, *dimensionless*
ρ	Density of fluid, $kg\ m^{-3}$
τ	Deviatoric stress tensor, *dimensionless*
τ_o	Fluid yield stress, Pa
ϕ	Angular position of the inner cylinder, *degrees*

Subscripts

max	Maximum value
BV	Bi-viscosity model
BE	Bercovier Engelman method

briefly described here.

Mack and Bishop [12] were seemingly the first to analytically study the steady, two-dimensional free convection in Newtonian fluids confined in the annular region between the two differentially heated isothermal concentric horizontal cylinders. They employed the power series expansion in Rayleigh number for stream function and temperature. The resulting streamline patterns for gases ($Pr \sim 0.7$) and non-metallic liquids were similar to the photographic results reported in the literature [13,14]. A single recirculating cell of the "crescent-eddy" shape was predicted and observed under the conditions spanned in their study ($Ra = 3000$ and $Pr = 0.7$). Further examination of their results reveals that though the role of Prandtl number is accentuated at its low values, the effect of Rayleigh number on the Nusselt number appears only in higher-order terms of their series solution, namely, at $O(Ra^4)$. Based on the first three terms of their series expansion, Mack and Bishop [12] also identified the limiting range of the Rayleigh number for the validity of their solution. In the range $1.15 \leq (R_o/R_i) \leq 4.15$ and $0.02 \leq Pr \leq 6 \times 10^6$, such limiting values were found to rise rapidly with the increase in values of (R_o/R_i) to its maximum, then decreasing with the increasing value of (R_o/R_i). They also developed a relationship between the critical value of the ratio (R_o/R_i) corresponding to the occurrence of Ra_{max} and the Prandtl number.

The next comprehensive study in this field is the combined experimental and numerical work of Kuehn and Goldstein [1] for water and air in a concentric annulus spanning the range of Rayleigh number (defined using gap-width between cylinders as a length scale) of 10^2 to 10^5. The values for heat transfer coefficients (both local and average) obtained from their experimental and numerical techniques showed a good agreement. Subsequently, Kuehn and Goldstein [15] experimentally examined the effect of eccentric placement of the inner cylinder on the rate of convective heat transfer over the range of Rayleigh numbers 2.2×10^2 to 7.7×10^7. At a fixed Rayleigh number, the average Nusselt number decreased by $\approx 5\%$ as the inner cylinder moves up along the vertical axis ($\varepsilon = 0.67$) and increases by $\approx 10\%$ as the inner cylinder is placed below the mid-axis along the vertical axis ($\varepsilon = -0.67$) compared to the concentric annular configuration. This can be safely attributed to the fact that when the inner cylinder is moved downwards, a greater mass of fluid is set in motion, whereas, for the opposite arrangement, much of the fluid beneath the inner heated cylinder transfers heat only by conduction. Subsequently, Kuehn and Goldstein [16] numerically explored the influence of Prandtl number ($0.001 \leq Pr \leq 1000$) and aspect ratio ($1 < R_o/R_i < \infty$) on the heat transfer coefficient (local and average) for a concentric annulus at a fixed value of $Ra = 10^4$. They reported that the heat transfer coefficient showed only a weak dependence on the Prandtl number for $Pr > 1$, beyond that inherent in the definition of the Rayleigh number. Also, the aspect ratio was reported to modulate the conditions of the transition for the onset of free convective heat transfer in the annular region. Greater the aspect ratio, higher is the rate of heat transfer. Cho et al. [17] numerically examined the influence of eccentricity on the heat transfer characteristics in the range of $10^3 \leq Ra \leq 5 \times 10^4$, $Pr = 0.7$ (air), for different aspect ratios and different azimuthal angular placement of the inner cylinder, e.g., see Fig. 1 here. For a fixed value of aspect ratio (R_o/R_i), the average heat transfer coefficient increased with the downward shift of the heated cylinder inside the annulus. This can probably be ascribed to the increasing degree of asymmetry in the flow field which promotes fluid mixing and heat transfer.

The influence of mixed boundary conditions on the free convective heat transfer characteristics in a concentric annulus has been investigated numerically by Kumar [18] for the aspect ratios in the range of 1.2–10 and $10^2 \leq Ra \leq 10^7$. The constant heat flux was applied on the inner cylinder whereas the outer cylinder was isothermal. They reported a higher heat transfer rate for constant heat flux cases compared to the isothermal wall condition for the same temperature difference. Ho et al.

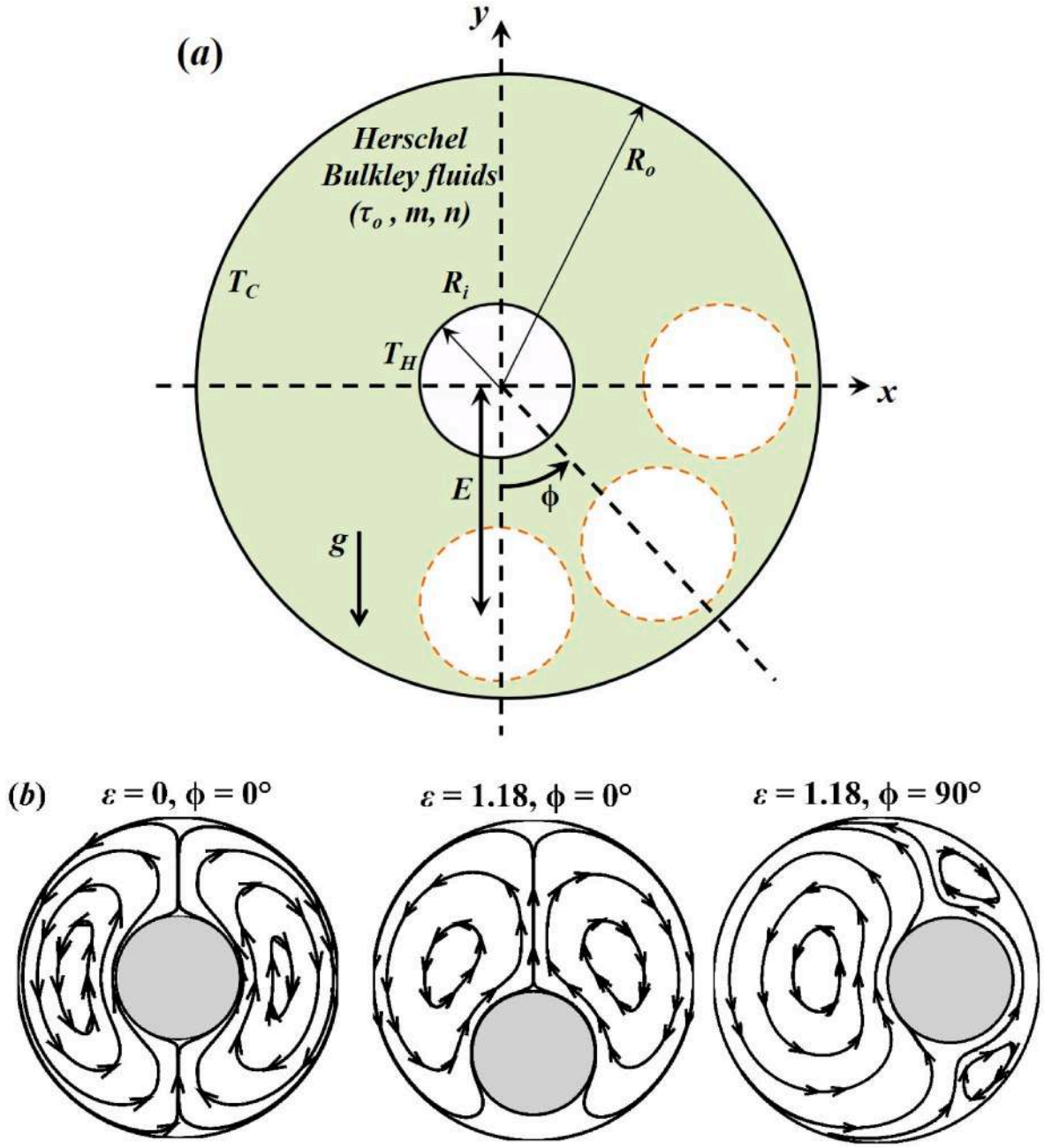


Fig. 1. Schematics of (a) the flow and computational domain, (b) flow domain showing the direction of the free convective flow currents.

[19] further examined the effect of eccentricity ($\varepsilon = \pm 0.625$) on heat transfer for $0.7 \leq Pr \leq 100$, $10^3 \leq Ra \leq 10^6$ and an aspect ratio of $R_o/R_i = 2.6$. They reported further enhancement in heat transfer by using the mixed boundary conditions. However, no general trend regarding the influence of eccentricity on the average Nusselt number could be established.

Similarly, numerous investigators have examined the transient free convection in a horizontal cylindrical annulus to establish the critical Grashof number for the transition of flow from the steady to unsteady regimes. Powe et al. [20] experimentally studied the free convective flow of air between two concentric isothermal cylinders. They reported that the critical Grashof number for the loss of steadiness of the flow showed a significant dependence on the relative gap between the two cylinders [$2R_i/(R_o - R_i)$]. They reported the critical Grashof number values of 10^5 and 1.55×10^5 for gap widths of 2 and 1.43, respectively. Later, Kuehn and Goldstein [15], Farouk and Guceri [21] and Tsui and Tremblay [22] all have examined the transient free convection in a concentric annulus and their results are consistent with the findings of Powe et al. [20]. Labonia and Guj [2] presented detailed experimental results on the flow transition characteristics and identified various flow regimes and transitions in a concentric annulus for Newtonian fluids. They reported the quasi-periodic regime to initiate at $Ra = 1.82 \times 10^5$ marking the loss of steadiness. Thereafter, several researchers have examined the multiple transitions and stability modes in transient free convection for Newtonian fluids in horizontal concentric and eccentric annuli [23–26]. In summary, significant literature exists on the laminar free convection heat transfer in concentric and eccentric horizontal annuli, at least for Newtonian fluids, though this body of knowledge is nowhere near as extensive and coherent as that for the free convection heat transfer from a single cylinder. Table 1 summarizes the ranges of parameters and conditions spanned by the pertinent studies available in this field.

Table 1
Summary of previous studies with Newtonian fluids.

Literature	Nature of study	Range of conditions
Kuehn and Goldstein [1]	Experimental and theoretical	$Ra = 24.41$ to 2.441×10^4, $R_o/R_i = 2.6$, $Pr = 0.7$, 7, Concentric
Labonia and Guj [2]	Experimental	$Ra = 2.2 \times 10^4$ to 8.22×10^4, $R_o/R_i = 2.36$, $Pr = 0.7$, Concentric
Kuehn and Goldstein [15]	Experimental	$Ra = 53.71$ to 1.879×10^7, $R_o/R_i = 2.6$, $Pr = 0.7$, 7, Eccentric
Kuehn and Goldstein [16]	Experimental	$Ra = 0.244 \times 10^4$, $0.001 \le Pr \le 10^3$, $1 < R_o/R_i < \infty$, Concentric
Cho et al. [17]	Numerical	$Ra = 0.244 \times 10^3$ to 1.22×10^7, $Pr = 0.7$, $R_o/R_i = 1.6$ to 5, Eccentric
Kumar [18]	Numerical	$Ra = 0.8$ to 7.29×10^9, $R_o/R_i = 1.2$ to 10, Concentric
Ho et al. [19]	Numerical	$Ra = 0.244 \times 10^3$ to 0.244×10^6, $0.7 \le Pr \le 100$, $R_o/R_i = 2.6$, Eccentric
Matin and Khan [33]	Numerical	$Ra = 3.375 \times 10^3$ to 3.375×10^5, $R_o/R_i = 2.5$, $10 \le Pr \le 10^3$, $0.6 \le n \le 1.4$, Concentric
Alsabery et al. [34]	Numerical	$Ra = 10^3$ to 10^6, $5 \le Pr \le 50$, $0.2 \le n \le 0.8$, $0.25 \le k_w \le 1.45$, Concentric
Masoumi et al. [35]	Numerical	$Ra = 0.125 \times 10^3$ to 0.125×10^8, $1.5 \le (R_o/R_i) \le 6$, $0 \le Y \le Y_{max}$, $Pr = 100$, Concentric

In contrast, despite the frequent occurrence of the different types of non-Newtonian fluid behaviour in a range of industrial settings [27–30], very few studies have examined the influence of non-Newtonian fluid characteristics on the free convective heat transfer in this geometry. From a theoretical standpoint, shear-dependent viscosity, yield stress and visco-elasticity further add to the difficulty of achieving not only uniform heating but the convective heat transfer rate is also often considerably lower than that in Newtonian fluids even in tube flows [31,32]. This difficulty is often associated with two inter-related factors, namely, the relatively small Grashof number values and progressively increasing viscosity far away from the cylinder surface due to the diminishing rate of fluid deformation. This difficulty is further accentuated for viscoplastic systems exhibiting a yield stress. Therefore, the present work is concerned with the laminar free convective heat transfer to a yield-pseudoplastic (modelled here as the Herschel-Bulkley model [28]) in the concentric and eccentric annular regions formed by two horizontal circular cylinders. A brief summary of the previous scant studies is included here. Matin and Khan [33] numerically examined the steady-state flow and heat transfer in free convection regime in power-law fluids filled in between the two isothermally heated concentric horizontal cylinders ($R_o/R_i = 2.5$) for $10^3 \le Ra \le 10^5$, $0.6 \le n \le 1.4$ and $10 \le Pr \le 10^3$. They reported an improvement in the rate of heat transfer by 170% for shear-thinning fluids ($n = 0.6$) and a reduction of 43% for shear-thickening fluids ($n = 1.4$) when compared to the corresponding case of Newtonian fluids (at fixed values of the *Gr* and *Pr*). Recently, Alsabery et al. [34] examined the free convection characteristics in a horizontal concentric annulus filled with blood, modelled by the Carreau rheological model [27,28]. Extensive numerical results in terms of isotherms, streamlines, Nusselt number (local and average) are presented for the range of Rayleigh number ($10^3 \le Ra \le 10^6$), power-law index ($0.2 \le n \le 0.8$), Prandtl number ($5 \le Pr \le 50$) and the non-dimensional thermal conductivity of the solid wall ($0.25 \le k_w \le 1.45$), respectively. An increase in the wall thermal conductivity and low values of power-law index were found to promote circulation of the blood in the enclosure. It is, however, not clear that what values of the Carreau model parameter (λ^* in Eq. (10) in their paper) have been employed in their simulations. Masoumi et al. [35] investigated the laminar free convection characteristics for Bingham plastic fluids in concentric cylindrical annulus for Rayleigh number ($10^3 \le Ra \le 10^5$), gap-width ($1.5 \le (R_o/R_i) \le 6$), Yield number ($0 \le Y \le Y_{max}$) at $Pr = 100$. Their so-called yield number (Y) is related to the present Oldroyd number as $Y = Od[((R_o - R_i)/R_i)(1/Gr)]^{0.5}$. Their results show that the yield number (i.e., yield stress) has a deleterious influence on heat transfer due to the dominance of conduction in the unyielded regions.

It thus stands to reason that the lowering of convection due to yield stress, as reported by Masoumi et al. [35], can be compensated to some extent by introducing shear-thinning behaviour in the fluid [33]. Indeed, this conjecture is well borne out in the case of forced convection from a sphere [36,37]. This possibility is explored in this work. In addition, this study also endeavours to elucidate the influence of the varying levels of eccentricity along and off the vertical axis of the annulus and this will indirectly delineate the effect of asymmetric flow field in the annulus. These objectives are accomplished here by obtaining and analysing extensive numerical results covering wide ranges of conditions as: Rayleigh number $10^3 \le Ra \le 10^5$, Oldroyd number $0 \le Od \le Od_{max}$, power-law index $0.2 \le n \le 1$, Prandtl number $10 \le Pr \le 100$, eccentricity $0 \le \varepsilon \le 1.18$ and angular position of the inner cylinder $0° \le \phi \le 180°$, for a fixed value of $(R_o/R_i) = 2.6$ in order to facilitate comparisons with previous studies [1,15,17,19]. Though bulk of the numerical results reported herein are based on the assumption of the steady flow, limited transient simulations have also been included here to establish the conditions for the loss of steady flow conditions to place these results in a proper context.

2. Problem definition

Consider two-dimensional free convective heat transfer in a yield-pseudoplastic fluid inside a horizontal circular annulus formed by two long horizontal cylinders of radius R_i (inner cylinder) and R_o (outer cylinder), Fig. 1. A constant value of the ratio of the two radii $R_o/R_i = 2.6$ is used here. The surface of the inner cylinder temperature is maintained at T_H, while the surrounding fluid and the outer cylinder are held at a lower temperature T_C ($<T_H$). The temperature difference induced density variation in the fluid gives rise to buoyancy-induced current in the annular fluid region, as shown schematically in Fig. 1 (b). The simplifying assumptions made here include steady, incompressible and laminar flow conditions. Considering the small temperature difference used here, $(T_H - T_C) = 5$ K, the fluid thermo-physical properties (τ_o, k, C_P, n and m) are assumed to be constant with respect to the temperature. Indeed, even the density is evaluated at the reference temperature (T_C) with an exception in the body force term of the momentum equation where the standard Boussinesq approximation is employed [38]. Similarly, the Brinkman number is very small (of the order of 10^{-3} (<1)) and hence viscous dissipation term can be safely neglected in the energy equation.

In order to examine the effect of annular eccentricity on the flow and heat transfer phenomena, the centre of the inner cylinder is shifted from its concentric location ($\varepsilon = 0$), Fig. 1, thereby constituting three different eccentric positions at $\varepsilon = 0.5$, 0.7 and 1.18 along the vertical axis where eccentricity ε ($= E/R_i$) is the non-dimensional distance between the centre of the inner and outer cylinders. Also, the angular position of the cylinder with respect to the vertical centre line ($\phi = 0°$, 45°, 90°, 135°, 180°) is varied to study the influence of asymmetric positioning of the heated inner cylinder inside the circular annular enclosure (Fig. 1(b)). Thus, altogether ten combinations of geometrical arrangements (values of ϕ and ε) of concentric and eccentric annuli have been studied here.

Under these conditions, the dimensionless field equations governing the coupled flow and temperature fields are given as follows:

$$\textbf{\textit{Continuity equation}}: \nabla . \boldsymbol{V} = 0 \tag{1}$$

$$\textbf{\textit{Momentum equation}}: \frac{D\boldsymbol{V}}{Dt} = -\nabla p + \sqrt{\frac{Pr}{Ra}}(\nabla . \boldsymbol{\tau}) + \Theta\, \delta_{ij} \tag{2}$$

$$\textbf{\textit{Energy equation}}: \frac{D\Theta}{Dt} = \frac{1}{\sqrt{(Pr.Ra)}}\nabla^2\Theta \tag{3}$$

For the steady flow regime, $\frac{\partial V}{\partial t} = 0$, $\frac{\partial \Theta}{\partial t} = 0$ therefore, the left-hand sides of Eq. (2) and Eq. (3) reduce to $\frac{DV}{Dt} = (\boldsymbol{V}.\nabla)\boldsymbol{V}$ and $\frac{D\Theta}{Dt} = (\boldsymbol{V}.\nabla)\Theta$, respectively.

Under incompressible flow conditions, the extra stress tensor (τ) is expressed as a function of the rate of deformation tensor as follows,

$$\boldsymbol{\tau}_{ij} = 2\eta\,\dot{\boldsymbol{\gamma}}_{ij} \tag{4}$$

where,

$$\dot{\boldsymbol{\gamma}}_{ij} = \frac{1}{2}\left[(\nabla \boldsymbol{V}) + (\nabla \boldsymbol{V})^T\right] \tag{5}$$

For a Herschel-Bulkley model fluid, the relation can be further expressed in terms of the Oldroyd number and power-law index as follows,

$$\boldsymbol{\tau} = \left(|\dot{\boldsymbol{\gamma}}|^{n-1} + \frac{Od}{|\dot{\boldsymbol{\gamma}}|}\right)\dot{\boldsymbol{\gamma}} \text{ if } |\boldsymbol{\tau}| > Od \tag{6a}$$

$$\dot{\boldsymbol{\gamma}} = 0 \qquad \text{if } |\boldsymbol{\tau}| \le Od \tag{6b}$$

The above equation approximates the rheological behaviour of viscoplastic fluids in the two regimes (solid-like and fluid-like). Thus, as long as the applied stress is lower than the fluid yield stress, it does not deform and behaves like an elastic solid, though this model does not account for elastic deformation in this region. As the stress level exceeds the fluid yield stress, such a material exhibits power-law fluid behaviour. It is important to note here that for $n = 1$, Eq. (6) shows the limiting case of the Bingham viscosity model. Thus, for $n = 1$ the Oldroyd number reduces to the familiar Bingham number. Furthermore, since Eq. (6) is non-differentiable and discontinuous at $\dot{\boldsymbol{\gamma}} = 0$, it cannot be employed as such into a numerical simulation methodology. This difficulty is surmounted by using one of the several regularization methods available in the literature which afford simple and effective way of numerically computing viscoplastic fluid flows [39–43]. The three commonly used regularization schemes of Papanastasiou [40], bi-viscosity model [41] and Bercovier and Engelman method [42] used here are presented below. The exponential model [40], reformulates the scalar viscosity function as follows,

$$\eta_P = 1 + \frac{Od\left[|\dot{\boldsymbol{\gamma}}|^{n-1} - exp\,(-M_P|\dot{\boldsymbol{\gamma}}|)\right]}{|\dot{\boldsymbol{\gamma}}|} \tag{7}$$

In Eq. (7), M_P is the dimensionless regularization parameter. It smoothens the abrupt transition from the solid-like behaviour to the fluid-like behaviour. In the limit of $M_P \to \infty$, Eq. (7) coincides with the Herschel-Bulkley viscosity model, Eq. (6). While very large values of M_P can lead to numerical instabilities, very small values of M_P (i.e., $M_P \to 0$) correspond to the power-law fluid behaviour and thus do not capture the yield stress effects adequately. Hence, an appropriate value of M_P is required for the accuracy of the numerical results. Similarly, the bi-viscosity model [41] is expressed as follows,

$$\eta_{BV} = |\dot{\boldsymbol{\gamma}}|^{n-1} + \frac{Od}{|\dot{\boldsymbol{\gamma}}|}\left(1 - \frac{\mu_{HB}}{\mu_Y}\right) \text{if } |\boldsymbol{\tau}| > Od \tag{8a}$$

$$\eta_{BV} = \left(\frac{\mu_Y}{\mu_{HB}}\right) \text{if } |\boldsymbol{\tau}| \le Od \tag{8b}$$

The solid-like regime is simulated here by a hypothetical fluid of very high yielding viscosity, μ_y, which is greater than the viscoplastic fluid viscosity, μ_{HB} by several orders of magnitude. Thus, a large value of μ_y/μ_{HB} is necessary to approach the true Herschel-Bulkley constitutive equation, but excessively large values can lead to the stiffness of the coefficient matrix thus resulting in potential convergence issues. Finally, the idea of Bercovier and Engelman [42] is expressed as follows,

$$\eta_{BE} = |\dot{\boldsymbol{\gamma}}|^{n-1} + \frac{Od}{|\dot{\boldsymbol{\gamma}}| + \lambda_{BE}} \tag{9}$$

The regularization parameter λ_{BE}, should be as small as possible for a good approximation of the true Herschel-Bulkley behaviour. As the value of λ_{BE} tends to zero, the ideal Herschel-Bulkley model is recovered. Though most of the results included are obtained by employing Eq. (7), the bi-viscosity and Bercovier and Engelman regularization schemes have been used for a few cases to perform a comparative analysis.

The standard boundary conditions for the present flow are written as follows,

Inner cylinder surface: No-slip ($\boldsymbol{V} = 0$) and constant temperature conditions ($\Theta = 1$).

Outer cylinder surface: No-slip ($\boldsymbol{V} = 0$) and isothermal temperature conditions of $\Theta = 0$.

The scaling variables used here are R_i for length, $V_C \sim \sqrt{R_i g\beta\,\Delta T}$for velocity, $m(V_c/R_i)^{n-1}$ for viscosity and $m(V_c/R_i)^n$ for the stress components, respectively. The non-dimensional temperature (Θ) is defined as $\Theta = (T - T_C)/(T_H - T_C)$. Evidently, for a fixed eccentricity (ε) and angular position (ϕ), the velocity and temperature fields are determined by the following four non-dimensional groups,

Rayleigh number:

$$Ra = \frac{(g\beta\,\Delta T)^{(3-n)/2} R_i^{(5+n)/2}}{(m/\rho)\alpha} = Gr.Pr \tag{10}$$

where Gr, the Grashof number, is defined as

$$Gr = \left(\frac{\rho}{m}\right)^2 (g\beta\,\Delta T)^{2-n} R_i^{2+n} \tag{11}$$

Prandtl number:

$$Pr = \left(\frac{mC_P}{k}\right)\left(\frac{g\beta\,\Delta T}{R_i}\right)^{\frac{n-1}{2}} \tag{12}$$

Oldroyd number:

$$Od = \frac{\tau_o}{m}\left(\frac{R_i}{g\beta\,\Delta T}\right)^{\frac{n}{2}} \tag{13}$$

The primitive variables, namely pressure (p), velocity ($\boldsymbol{V}$) and temperature (T) obtained by the numerical solution of the above-mentioned governing equations, are further processed to determine the local and average quantities like streamline, isotherms, structure of yield surfaces, Nusselt number (local and average), as detailed elsewhere [6,44]. The gross parameters employed here are described in brief as follows,

Drag coefficient (C_D):

$$C_D = \frac{2F_D}{\rho V_c^2 (2R_i)} \tag{14}$$

where, F_D is the total drag force per unit length experienced by the inner cylinder.

Local Nusselt number (Nu_θ):

$$Nu_\theta = -\left.\frac{\partial \Theta}{\partial n_s}\right|_{surface} \tag{15}$$

where, n_s is the unit normal vector (outward) on the inner cylinder surface.

Average Nusselt number (Nu_{avg}):

$$Nu_{avg} = \frac{1}{A_s}\int_{A_s} Nu_\theta dA_s \tag{16}$$

where, A_s is the surface area of the inner cylinder.

Though the drag coefficient is not a quantity of direct interest here, it is employed herein only for making a suitable selection of the numerical parameters, i.e., grid size and regularization parameter, convergence, etc.

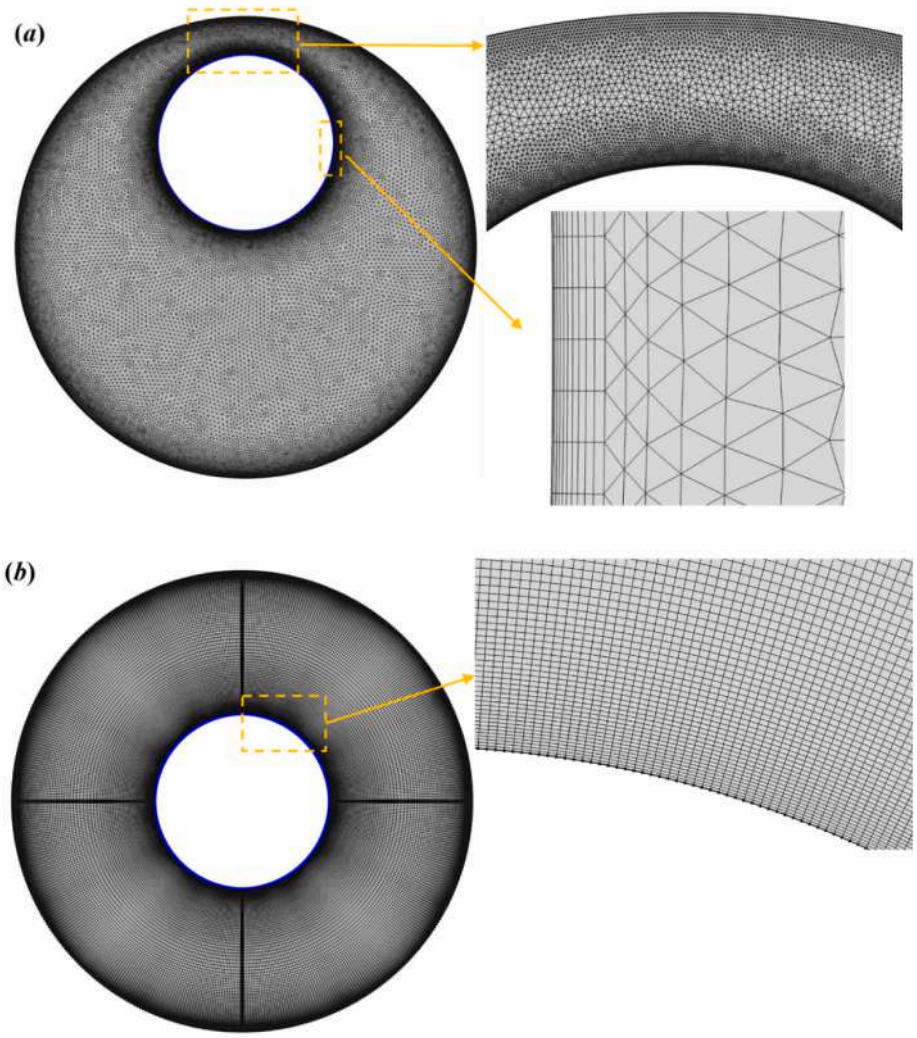


Fig. 2. Schematic representation of a typical grid structure for (a) $\varepsilon = 1.18$, $\phi = 180^{\circ}$ and (b) $\varepsilon = 0$, $\phi = 0^{\circ}$.

3. Numerical scheme

The preceding governing equations (Eqs. (1)–(3)) along with the boundary conditions and the Herschel-Bulkley fluid model have been solved by using the finite element-based solver COMSOL Multiphysics (Version 4.3a). The volumetric source term (buoyancy force) and the Herschel-Bulkley viscosity relation have been incorporated into the numerical scheme via user defined functions (UDFs). The built-in CAD tools and meshing functions are employed for creating the flow geometry and the computational grid structure. Due to the expected sharp gradients close to the inner cylinder surface, this region was meshed using boundary layers (10 nos.) with a constant stretching factor of 1.2. Beyond this region, the computational domain was meshed using non-uniform free triangular and free-quadrilateral grids. Also, symmetric grid distributions have been used herein to ensure a fine grid framework close to the outer cylinder surface. Based on a trial-and-error approach,

Table 2
Grid-independence test at $Ra = 10^5$, $\varepsilon = 0$, $\phi = 0^{\circ}$.

Pr	Grid	N_E	N_S	$n = 0.2$			$n = 1$		
				C_d	Nu_{avg}	ξ	C_d	Nu_{avg}	ξ
***Od* = 0.01**									
10	*M1*	55,100	200	−0.7260	6.2785	0.9907	−0.7087	4.9801	0.9829
	M2	92,500	300	−0.7407	6.4236	1	−0.7377	5.0772	0.9961
	M3	142,200	350	−0.7411	6.4231	1	−0.7381	5.0775	0.9967
100	*M1*	55,100	200	−0.5856	3.6983	0.9951	−0.7623	4.7819	0.9934
	M2	92,500	300	−0.6006	3.7375	1	−0.7757	5.0524	0.9935
	M3	142,200	350	−0.6007	3.7376	1	−0.7757	5.0524	0.9937
***Od* = 2**									
10	*M1*	55,100	200	−0.4577	2.9802	0.9256	−0.4713	3.0518	0.8365
	M2	92,500	300	−0.4816	3.2023	0.9490	−0.5047	3.2397	0.8537
	M3	142,200	350	−0.4816	3.2024	0.9487	−0.5046	3.2397	0.8535
100	*M1*	55,100	200	−0.3014	1.0417	0.0817	−0.3295	1.0447	0.0081
	M2	92,500	300	−0.3133	1.0451	0.0828	−0.3311	1.0451	0.0087
	M3	142,200	350	−0.3135	1.0451	0.0829	−0.3312	1.0451	0.0087

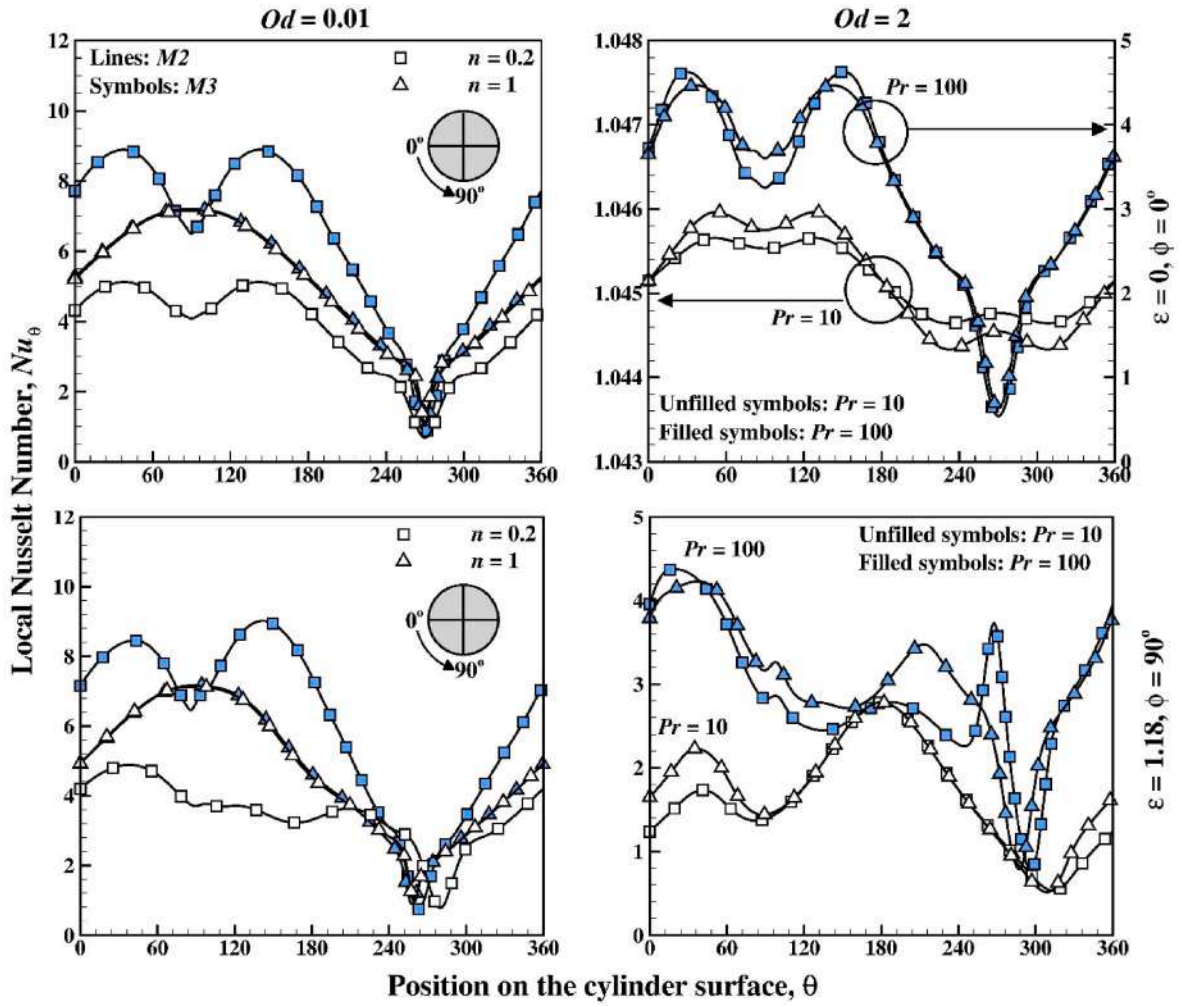


Fig. 3. Influence of grid resolution on the local Nusselt number at $Ra = 10^5$, $Pr = 10$, 100, $Od = 1$, for (a) $\varepsilon = 0$, $\phi = 0°$ and (b) $\varepsilon = 1.18$, $\phi = 90°$.

the relative convergence criterion of 10^{-5} for each primitive variable was found to be adequate for the present work. Also, the relative tolerance of 10^{-5} for the von-Mises yielding criterion [45] has been employed herein to demarcate the apparent yield surfaces. This means that the magnitude of the local stress tensor and the fluid yield stress at any point are within 0.001% to determine the state of the material, i.e., fluid-like yielded zone or solid-like unyielded zone.

Extensive grid sensitivity tests have been carried out here for each geometry (i.e., pair of values of ε and ϕ) at $Ra = 10^5$, $Od = 0.01$, 2, $n = 0.2$, 1 and $Pr = 10$, 100. Fig. 2 shows the typical grid structure for $\varepsilon = 0$, $\phi = 0°$ and $\varepsilon = 1.18$, $\phi = 180°$, used in the present work. Grids (*M1*, *M2*, and *M3*) of varying mesh density and quality were employed for this purpose. Table 2 presents a succinct summary of the results of grid independence tests for $\varepsilon = 0$, $\phi = 0°$. Clearly, the results vary by less than 1% as the grid size is refined from *M2* to *M3*. Hence, grid *M2* is suitable to capture the sharp gradients under these conditions. Fig. 3 shows the effect of numerical grid in terms of the local Nusselt number (Nu_θ) at $Ra = 10^5$, $Pr = 10$, 100, $Od = 0.01$, 2 and two combinations of ε and ϕ ($\varepsilon = 0$, $\phi = 0°$, $\varepsilon = 1.18$, $\phi = 90°$). Clearly, the profiles obtained with grid *M2* and *M3* overlap, further justifying the choice of grid *M2*. In addition, the apparent yield surfaces were also compared for each mesh (*M1*, *M2* and *M3*) and grid *M2* was found to adequately resolve the apparent yield surfaces. Similarly, the mesh was refined close to the surface of the outer cylinder. The Nu_{avg} for both the cylinders have been compared at high Rayleigh number ($Ra = 10^5$) for grid *M2*. The two values differ by less than 1%, thus lending further support to the present choice of the numerical mesh.

Table 3
Influence of regularization parameter (M_P) for $Ra = 10^5$, $Pr = 10$, $Od = 2$.

M_P	$n = 0.2$			$n = 1$		
	C_d	Nu_{avg}	ξ	C_d	Nu_{avg}	ξ
$\varepsilon = 0$, $\phi = 0°$						
10^5	−0.4789	3.2017	0.9411	−0.5032	3.2386	0.8517
10^6	−0.4816	3.2021	0.9487	−0.5044	3.2393	0.8535
10^7	−0.4821	3.2024	0.9489	−0.5047	3.2395	0.8539
$\varepsilon = 1.18$, $\phi = 90°$						
10^5	−0.7447	2.9063	0.9189	0.6934	3.0508	0.8593
10^6	−0.7461	2.9099	0.9238	0.6951	3.0533	0.8765
10^7	−0.7465	2.9102	0.9242	0.6957	3.0536	0.8769

Table 3 shows the effect of the regularization parameter (M_P) on the drag coefficient, average Nusselt number, the fraction of yielded zones and the structure of apparent yield surfaces in the entire flow domain at $Ra = 10^5$, $Pr = 10$, $Od = 2$. The use of values of $M_P > 10^6$ was found to be computationally highly intensive with very little change in the results. Hence the value of $M_P = 10^6$ is appropriate for the range of conditions spanned herein. These predictions are also corroborated by employing the other regularization approaches given by Eq. (8) and Eq. (9). A close agreement exists between the three predictions. However, for the sake of brevity, these results have not been included here. Overall, the present numerical results are based on grid *M2*, $M_P = 10^6$ and convergence criterion of 10^{-5}. Further justification for this choice of numerical parameters and the numerical methodology is presented by reporting a few

Table 4
Comparison with the results of Kuehn and Goldstein [1] for mean equivalent conductivity (k_{eq}) on the inner and outer cylinders at $Pr = 5$, $(R_o\text{-}R_i)/R_i = 1.6$ for a concentric annulus.

Ra	Kuehn and Goldstein [1]		Present results	
	k_{eq} (inner)	k_{eq} (outer)	k_{eq} (inner)	k_{eq} (outer)
10^4	2.069	2.066	2.036	2.039
5×10^4	3.036	3.088	2.972	2.977
10^5	3.756	3.471	3.481	3.487

benchmark validations with the previous results in the literature for Newtonian and non-Newtonian fluids.

Few transient simulations have also been done to ascertain the critical values of the Rayleigh number for the cessation of the steady flow regime. A small time step was employed in the Generalized alpha time-integration scheme (implicit, second-order accurate), along with the above-mentioned parameters, namely grid size, value of the regularization parameter, etc. at $Ra = 10^6$ regardless of the values of Pr, Od, n, ε (eccentricity) and ϕ (angular placement of the inner cylinder).

4. Results and discussion

4.1. Validations

As wide-ranging comparisons spanning Newtonian, Bingham plastic and power-law fluids for laminar free convection in a square cavity with and without a heated cylinder have already been reported earlier [6,36,37], only additional comparisons for the case of concentric and eccentric annuli are presented here. Table 4 contrasts the present values with the numerical results of Kuehn and Goldstein [1] in terms of the mean conductivity (k_{eq}) on both cylinder surfaces of the concentric annulus at $Pr = 5$ and $(R_o/R_i) = 2.6$. The mean conductivity is defined here as the ratio of the rate of heat transfer by convection to that by conduction. Generally, the two predictions are close to each other, but the deviations of up to 7% seen here can be attributed to the differences in the numerical scheme and the relatively coarser mesh used in ref. 1. Fig. 4 shows a comparison in terms of the temperature and velocity variation along the horizontal midline at $Pr = 100$, $Ra = 10^4$, 10^6, $R_o/R_i = 3$ and for three values of the yield number (Y) with the work of Masoumi et al. [35]. The two predictions almost superimpose exhibiting good

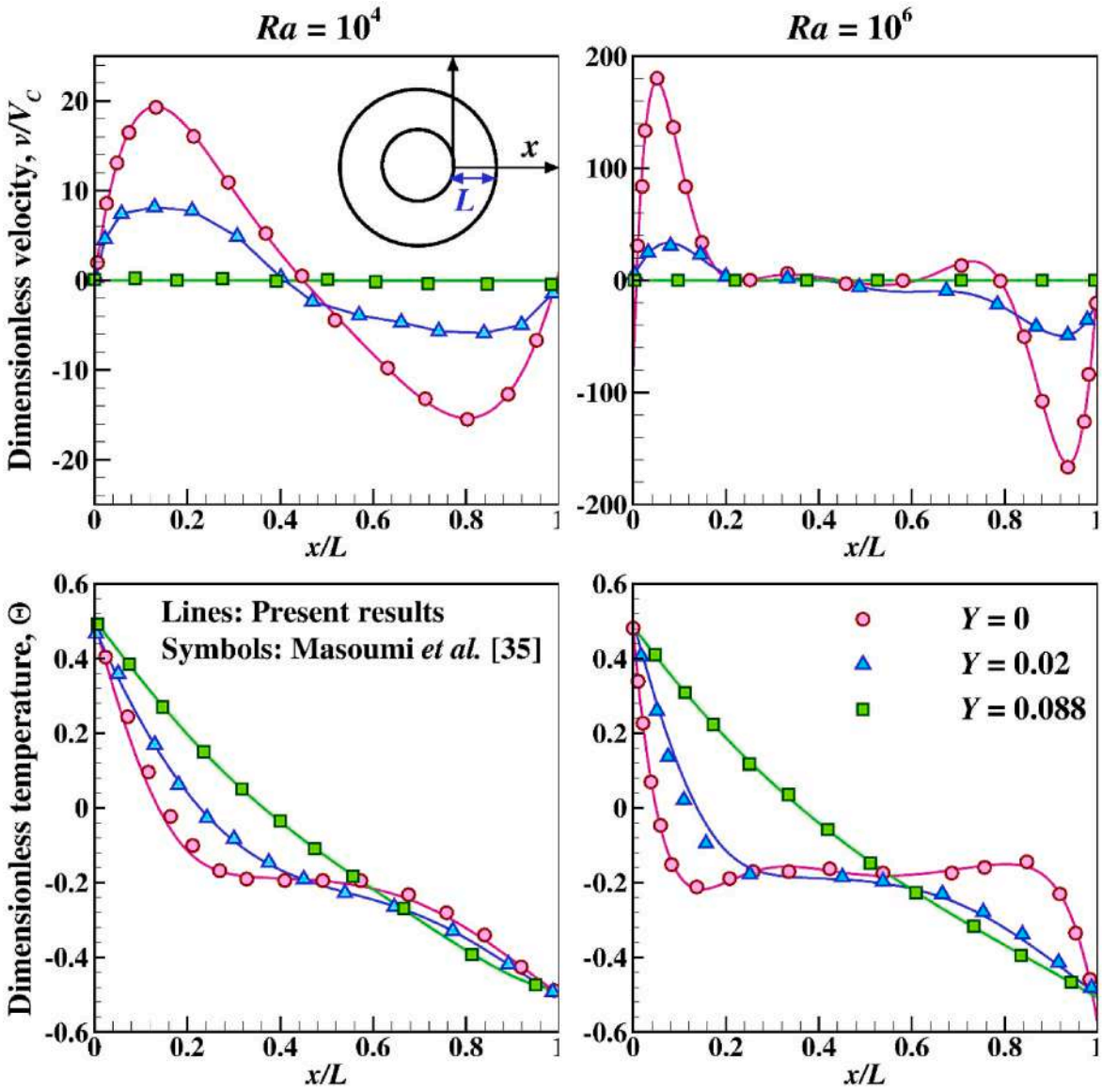
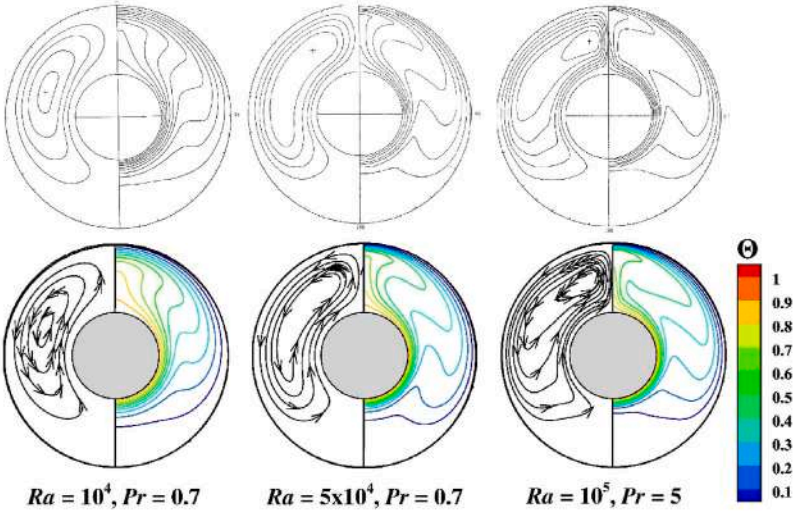
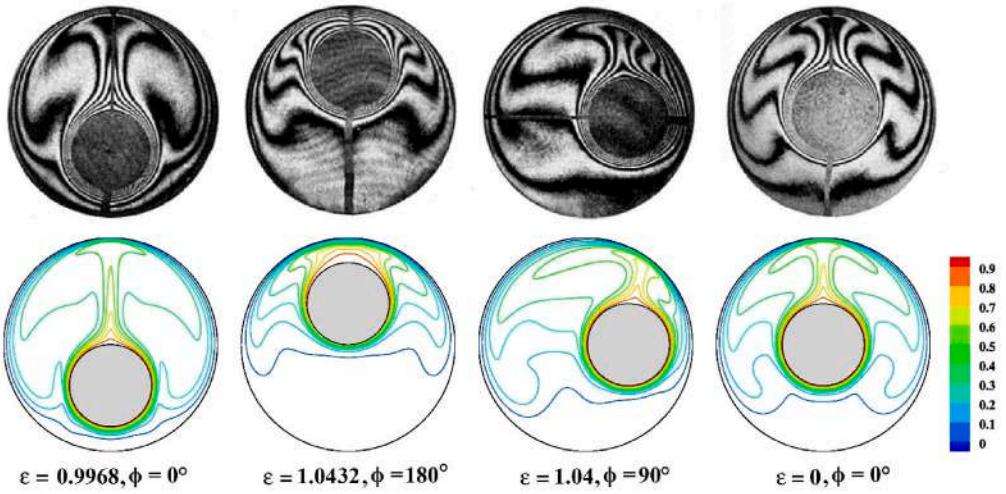


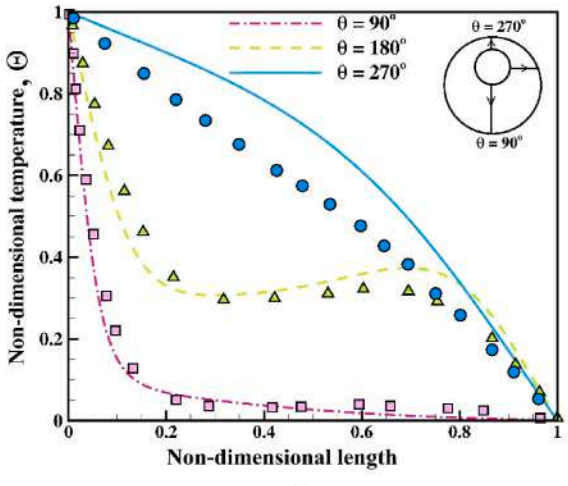
Fig. 4. Comparison of the present results of non-dimensional velocity (v) and temperature (Θ) along the cross-section of the concentric annulus with the results of Masoumi et al. [35] at $Ra = 10^4$, 10^6, $Pr = 100$, $R_o/R_i = 3$ and different values of yield number (Y).



a



b



c

Fig. 5. (a): Comparisons of stream functions ($\Delta\psi = 0.5$) and isotherm contours ($\Delta\Theta = 0.1$) with the experimental results of Kuehn and Goldstein [1] and present results at (R_o/R_i) = 2.6 and different values of Rayleigh and Prandtl numbers. b: Comparisons of isotherm contours isotherm contours ($\Delta\Theta = 0.1$) with the experimental results of Kuehn and Goldstein [15] and present results at $Ra = 4.8 \times 10^4$, $Pr = 0.7$, (R_o/R_i) = 2.6. cNon-dimensional temperature variation along the non-dimensional length (along the vertical/horizontal central axis) at $Ra = 4.8 \times 10^4$, $Pr = 0.7$, $\varepsilon = 1.0432$, (R_o/R_i) = 2.6. Symbols show experimental results of Kuehn and Goldstein [15] and lines represent present results.

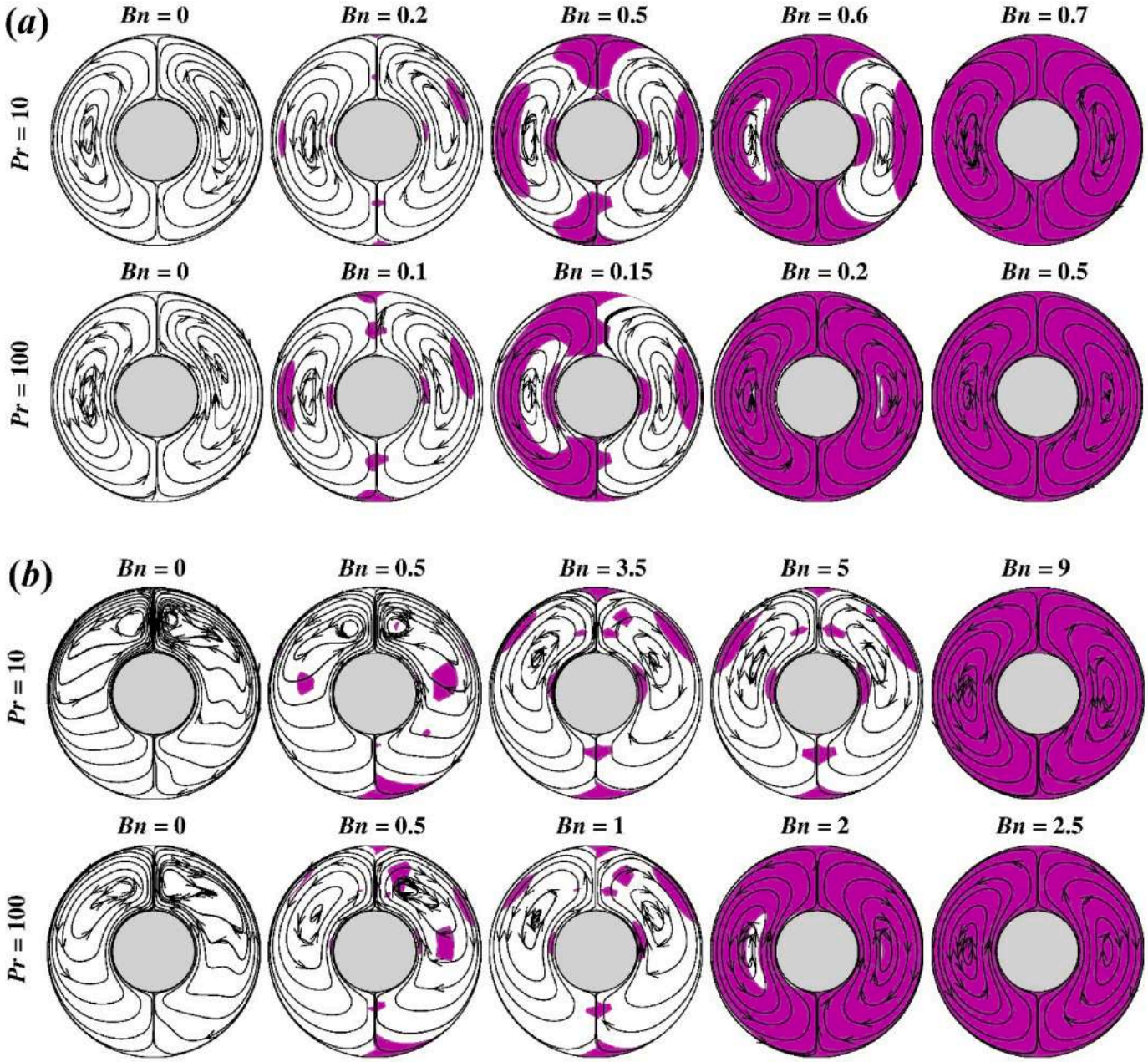


Fig. 6. Streamlines and yield surfaces at $\varepsilon = 0$, $\phi = 0^\circ$ for $n = 0.2$ (Left half), $n = 1$ (Right half) at (a) $Ra = 10^3$, (b) $Ra = 10^5$. (: Unyielded zone,: Yielded zone).

correspondence. Finally, the experimental predictions of Kuehn and Goldstein [15] in terms of stream function, isotherms, and temperature in Newtonian fluids are compared in Fig. 5a–c, respectively. In most cases, the agreement is seen to be fair, the deviations being well within the experimental uncertainty. Keeping in view, the preceding comparisons and those reported elsewhere [6,36,37], the present results are considered to be reliable within ± 2–3%.

4.2. Morphology of yielded/unyielded zones

Figs. 6 and 7 show the influence of the relevant parameters (*Ra*, *Pr*, *n*, *Od*) on the size, shape and position of the apparent yield surfaces for $\varepsilon = 0$, $\phi = 0^\circ$ and $\varepsilon = 1.18$, $\phi = 0^\circ$. Evidently, a Herschel-Bulkley model fluid will yield only in the region where the prevailing stress exceeds the fluid yield stress. However, in view of the regularized forms of the Herschel-Bulkley model used here, the surface demarcating the unyielded and yielded zones must be seen as constant stress contours instead of being the true yield surfaces. The limiting case of $Od = 0$ (power-law fluid) is also included here for reference wherein no unyielded sub-domain are expected and observed. The unyielded regions are distributed throughout the flow domain in the form of small blobs of varying shapes and sizes. At a fixed Rayleigh number, the Grashof number increases with the decreasing Prandtl number. This promotes the buoyancy-induced flow and hence, large fluid-like regions are formed. For low Grashof numbers, shear-thinning viscoplastic fluid behaviour ($n < 1$) leads to smaller fluid-like regions when compared to that for Bingham plastic fluids ($n = 1$) at a fixed Bingham number or Oldroyd number. At high Grashof numbers, the buoyancy (or fluid inertia) dominates the flow and hence larger fluid-like regions are seen for shear-thinning fluids. This suggests an enhanced rate of heat transfer on account of shear-thinning viscosity at large Grashof numbers when there is a modest degree of convection present in the fluid. This is similar to the role of flow behaviour index seen in the context of forced convection in viscoplastic fluids from a heated sphere [36,37]. With the increasing

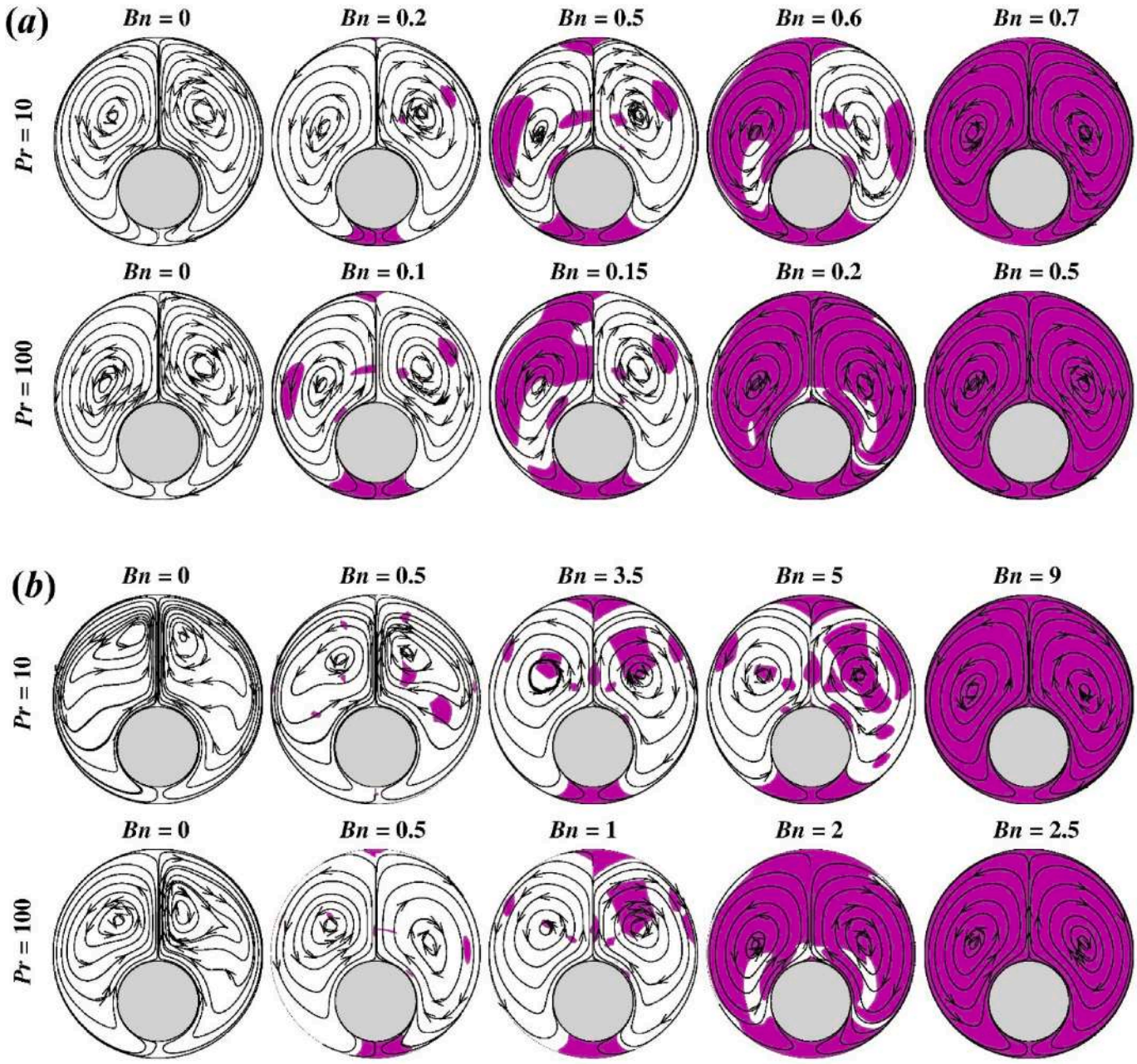


Fig. 7. Streamlines and yield surfaces at $\varepsilon = 1.18$, $\phi = 0^{\circ}$ for $n = 0.2$ (Left half), $n = 1$ (Right half) at (a) $Ra = 10^3$, (b) $Ra = 10^5$. (: Unyielded zone,: Yielded zone).

Oldroyd number, the influence of shear-dependent viscosity on the yield surfaces weakens and the enhanced yield stress effects suppress the buoyancy forces and the fluid-yielding, i.e., the unyielded (solid-like) regions grow under these conditions. It thus stands to reason that for a given geometric configuration (eccentricity, ε and angular placement of inner cylinder ϕ) and for each combination of *Gr*, *Pr* and *n*, there exists a limiting value of *Od* (Od_{max}), similar to the Yield parameter in Ref. [35], above which the flow ceases (or the fluid is frozen so to say) and conduction is the sole means of heat transfer. From another vantage point, this can be seen in Fig. 8 where the representative results on the fraction of the cross-section occupied by the fluid-like regions is plotted as a function of the influencing parameters. The results in the middle row of plots are particularly informative and show the role of eccentricity. Evidently the fluid yielding persists up to higher values of Oldroyd number in an eccentric annulus than that in a concentric annulus. This should promote convective heat transfer. This limit is also known as the fully plastic limit. Now returning to the question of the limiting value of the Oldroyd number corresponding to the fully plastic limit, it is the value of Oldroyd number at which the fraction of yielded zone (ξ) becomes equal to zero (Fig. 8). For a constant Rayleigh number, the Od_{max} decreases as the Prandtl number increases. The limiting Oldroyd number increases by almost a factor of 10–15 as the Rayleigh number is varied from 10^3 to 10^5 and therefore widening the operating envelope of such a system. The power-law index (*n*) seems to play little role above that accounted for by its appearance in the definitions of *Ra*, *Pr* and *Gr*. For low Grashof numbers, the yield-pseudoplastic fluids achieve the fully plastic limit quickly compared to the Bingham plastic fluids. This trend, however, reverses for Grashof numbers $\geq 10^3$. This behaviour can be explained by surmising that there is a negligible motion at low Grashof numbers and therefore, the flow properties like viscosity are of no consequence in the context of conduction.

The geometric factors (ε and ϕ values), also influence the structure of

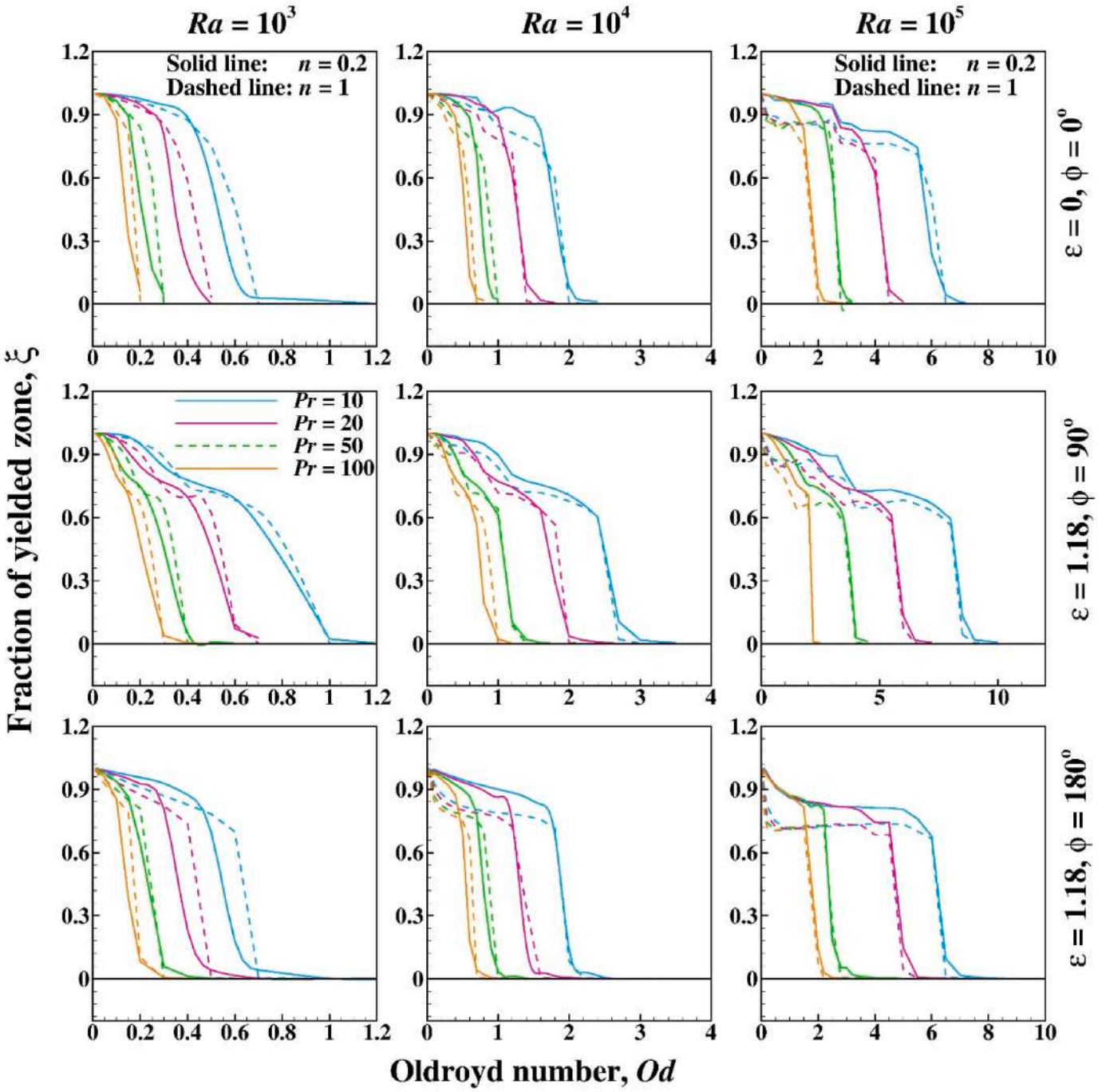


Fig. 8. Variation of the fraction of yielded zone with Bingham number for different values of *Ra*, *Pr*, *n*, and two different combinations of ε and ϕ.

the flow field, and hence the value of the Od_{max} to achieve the fully plastic conditions. For a concentric annulus ($\varepsilon = 0$, $\phi = 0°$), the unyielded zones grow symmetrically about both the *x*- and *y*-axis at low Rayleigh numbers such as $Ra = 10^3$. Solid-like zones are dispersed as small pockets at the axial locations (in the front, centre and in the rear of the cylinder). At high Rayleigh numbers, the unyielded regions are more scattered around the heated cylinder. At $\phi = 0°$ (symmetric placement), both the size of the yielded zones and the Od_{max} increase with the downward shift of the inner cylinder. At $\varepsilon = 1.18$, the inner cylinder is close to the outer cylinder wall and the small gap in between the two acts like a dead zone. Thus, unyielded zones are formed in the small intervening space even for very small values of Oldroyd number. Overall, the eccentric placement of the heated cylinder results in an increase in the maximum Oldroyd number values with reference to the case of a concentric annulus thereby expanding the envelope of operation if fluid yielding is desired in a given application. The horizontal shift of the inner cylinder, i.e., increasing value of ϕ (angular position) slows down the approach to the completely solid-like zone and this also broadens the range of operating conditions with significant fluid yielding. Hence, the limiting Oldroyd number is higher for $\phi = 90°$ than that at $\phi = 0°$ irrespective of the value of eccentricity (Fig. 8). The vertical shift feebly influences the values of maximum Oldroyd number. Hence, the fraction of yielded zone goes to zero nearly at the same values of Oldroyd number for $\varepsilon = 1.18$, $\phi = 180°$ as that observed for $\varepsilon = 0$, $\phi = 0°$.

4.3. Streamlines and isotherms

While Figs. 6 and 7 show typical streamline profiles for a range of

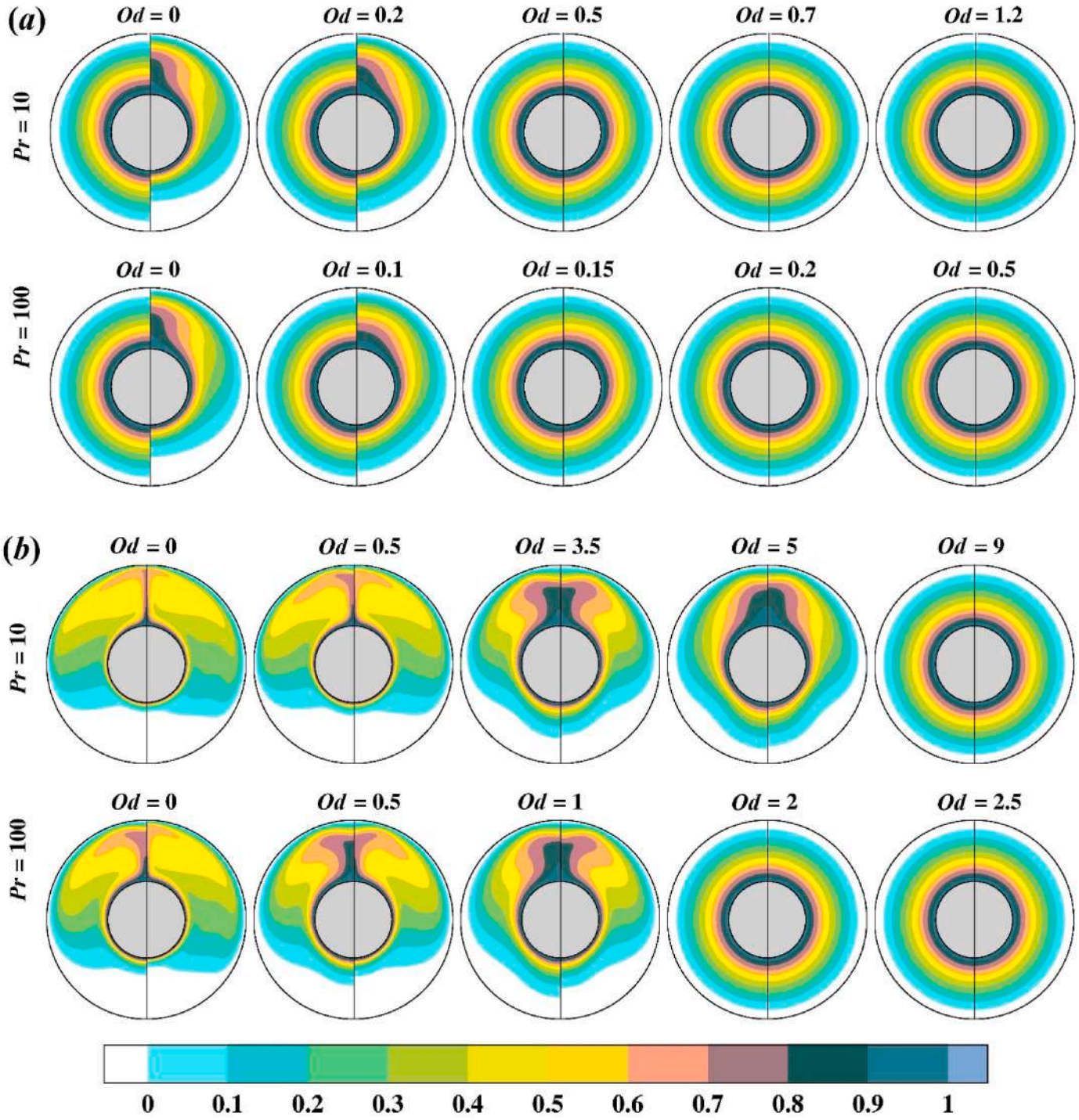


Fig. 9. Isotherm contours at $\varepsilon = 0$, $\phi = 0^{\circ}$ for $n = 0.2$ (Left half), $n = 1$ (Right half) at (a) $Ra = 10^3$, (b) $Ra = 10^5$.

conditions as discussed above, Figs. 9 and 10 show typical isotherm contours. We begin by examining the base case of a concentric annulus, i.e., $\varepsilon = 0$, $\phi = 0^{\circ}$ (Fig. 6). Owing to the symmetry of the flow, the results are plotted in the left half for $n = 0.2$ and the right half for $n = 1$. Two symmetric bean-shaped recirculating cells are observed at low Rayleigh numbers (10^3). For shear-thinning fluids, the fore and aft symmetry exists in the annulus signifying weak flow conditions. Under these conditions, the Prandtl number does not influence the flow patterns due to weak advection. As the Rayleigh number increases, strong recirculating zones are developed in the upper half of the annulus. However, the lower half of the annulus remains almost static with feeble circulatory patterns and thus little heat transfer occurs here. Further, the fluid yield stress imparts flow stability by suppressing the buoyancy effects. Also, the effect of power-law index on the streamlines subsides with an increase in the Oldroyd number as the yield stress forces overshadow the viscous effects.

Turning to the corresponding isotherms, Fig. 9 shows isotherms for a concentric annulus ($\varepsilon = 0$, $\phi = 0^{\circ}$). At low values of *Ra*, the flow velocity is too weak to influence the temperature field. Hence, the temperature contours follow the body contour of the heated inner cylinder akin to that in conduction. The assertions made above regarding the influence of the power-law index are also borne out by the isotherm contours shown

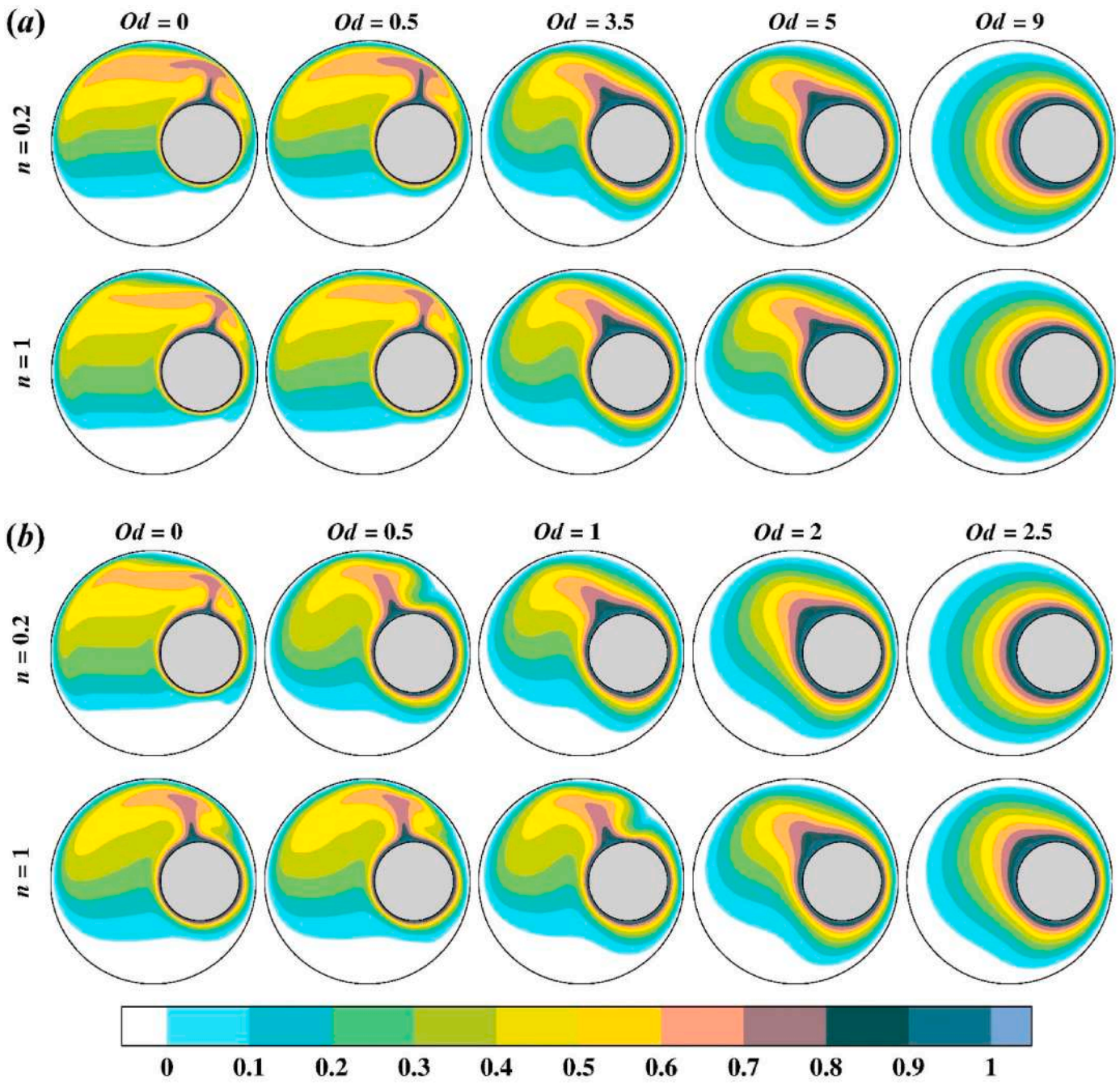


Fig. 10. Isotherm contours at $\varepsilon = 1.18$, $\phi = 90^\circ$ for $Ra = 10^5$, $n = 0.2$, 1 at (a) $Pr = 10$, (b) $Pr = 100$.

here. Clearly, a thermal plume is visible for $n = 1$ at $Od = 0$, $Ra = 10^3$ in contrast to the concentric circular temperature contours at $n = 0.2$ indicating the dominant role of conduction under these conditions. An upward rising thermal plume appears at large values of the Ra and Pr, owing to the progressive thinning of the momentum and thermal boundary layers. This phenomenon is counteracted by the increasing Oldroyd number. Further increase in Oldroyd number leads to typical symmetric, temperature distributions of the form of concentric circles, a clear signature of the conduction-dominated heat transfer mechanism.

Similar remarks can be made about the flow and temperature fields at $\varepsilon = 1.18$, $\phi = 0^\circ$. Due to the downward shift of the heated cylinder, the circulation is enhanced resulting in larger fluid entrainment and improved heat exchange between the cylinder and the fluid. For $\phi = 0^\circ$ as eccentricity (ε) increases, the vertical distance from the outer cold wall increases, allowing more space for the thermal plume to expand, mobilizing a larger fluid volume. Thus, the dead zones formed below the heated inner cylinder are significantly reduced, leading to an improved rate of heat transfer. For $\varepsilon = 1.18$, $\phi = 180^\circ$, the hot cylinder is very close to the cold outer cylinder. This close proximity decreases the expanse and strength of the convective plume thus indicating a reduced rate of convective heat transfer. For symmetric placement of the heated cylinder in the annulus ($\phi = 0^\circ$), streamlines remain symmetric about the y-axis up till $Ra = 10^5$ for the entire parameter range embraced here. Beyond $Ra = 10^5$, the flow transits to the transient flow behaviour and hence the symmetry (about the y-axis) is lost for some values of eccentricity (ε) used here. The range of conditions for the loss of flow symmetry and onset of transient flow characteristics will be discussed later.

For the asymmetric placement of the inner heated cylinder, two recirculation cells of different sizes and shapes are established. Also, for a constant value of eccentricity (ε), the fluid volume above the heated

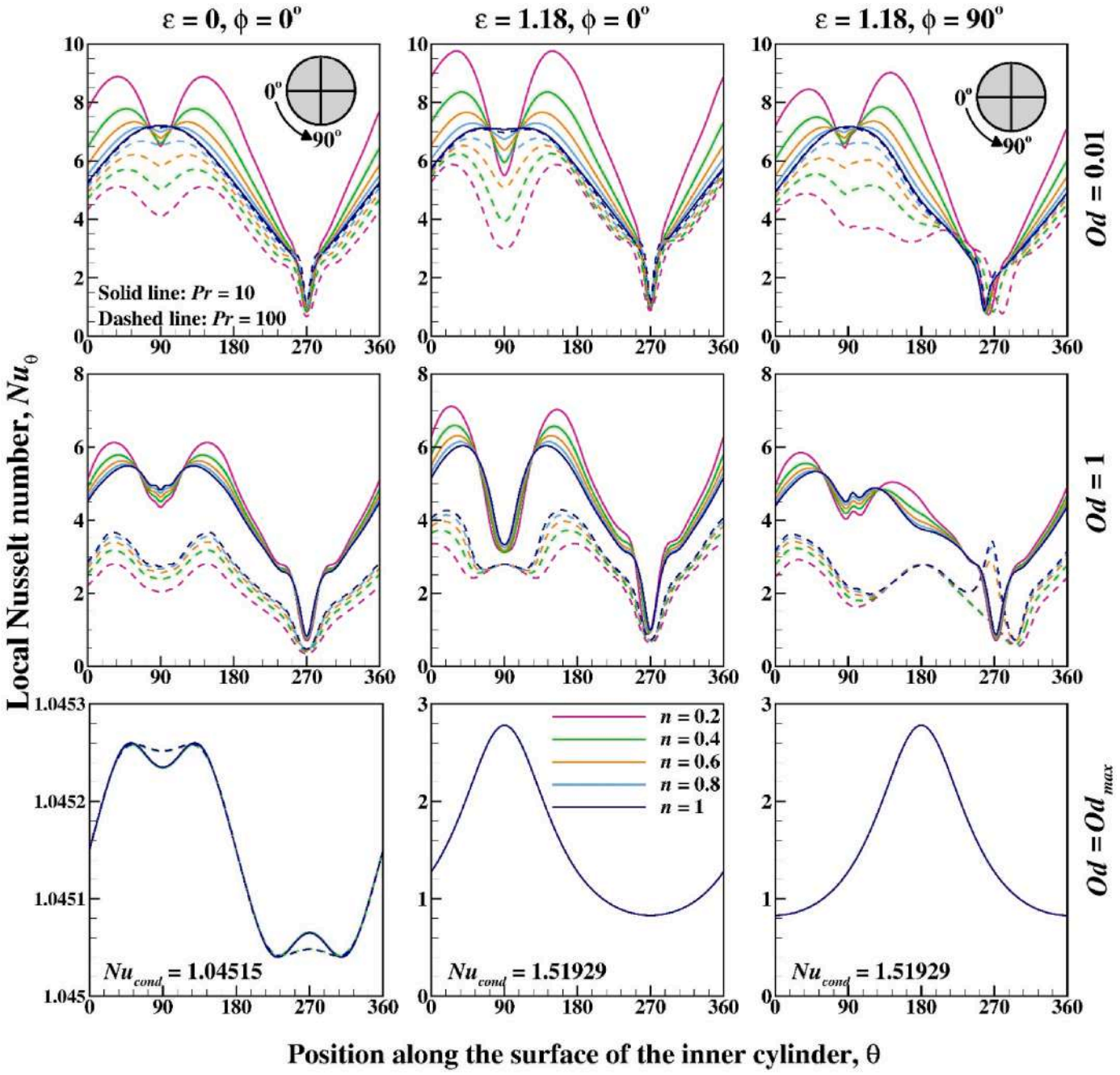


Fig. 11. Variation of local Nusselt number (Nu_θ) over the surface of the inner cylinder at $Ra = 10^5$, $Pr = 10$, 100 and different values of n, Od, ε and ϕ.

cylinder is restricted with an increase in the value of angular position (ϕ) and hence the intensity and size of the thermal plume is reduced. For the case of $\varepsilon = 1.18$, $\phi = 90^\circ$, the heated cylinder is in close proximity of the cold outer cylinder. These walls obstruct the ascent of the thermal plume. Correspondingly, the convective plume bends leftward forming a large primary recirculation cell engulfing bulk of the region in the annulus. This radial spread is at the cost of retardation of the rising buoyant plume thereby leading to a reduction in the convective heat transfer. Two small secondary vortices are also formed above and below the inner cylinder in the narrow space between the two circular cylinders. In the convection dominant regime, i.e., high value of Ra, the upper secondary plume is larger than the one formed below the cylinder. As the Oldroyd number increases, the flow profiles become increasingly symmetrical about the x-axis due to the diminishing role of convection. Overall, the strength of the buoyancy currents and the thermal plume are reduced for all values of eccentricity (ε) at $\phi = 90^\circ$ compared to the case of a concentric annulus. Additional insights can be obtained by looking at the variations in the local velocity and temperature. At $Ra = 10^5$ and low values of Oldroyd number, the magnitude of vertical velocity is higher for shear-thinning fluids than that in Newtonian fluids signifying an increased strength of convective currents. As Oldroyd number increases, the velocity decays sharply. However, the power-law index does not seem to influence the velocity profiles beyond $Od = 1$. This is also consistent with the streamline (Figs. 6 and 7) and isotherm profiles (Figs. 9 and 10) seen here.

4.4. Nusselt number

The overall performance for heat transport is gauged by the local and surface average Nusselt numbers. Typical results for the variation of the

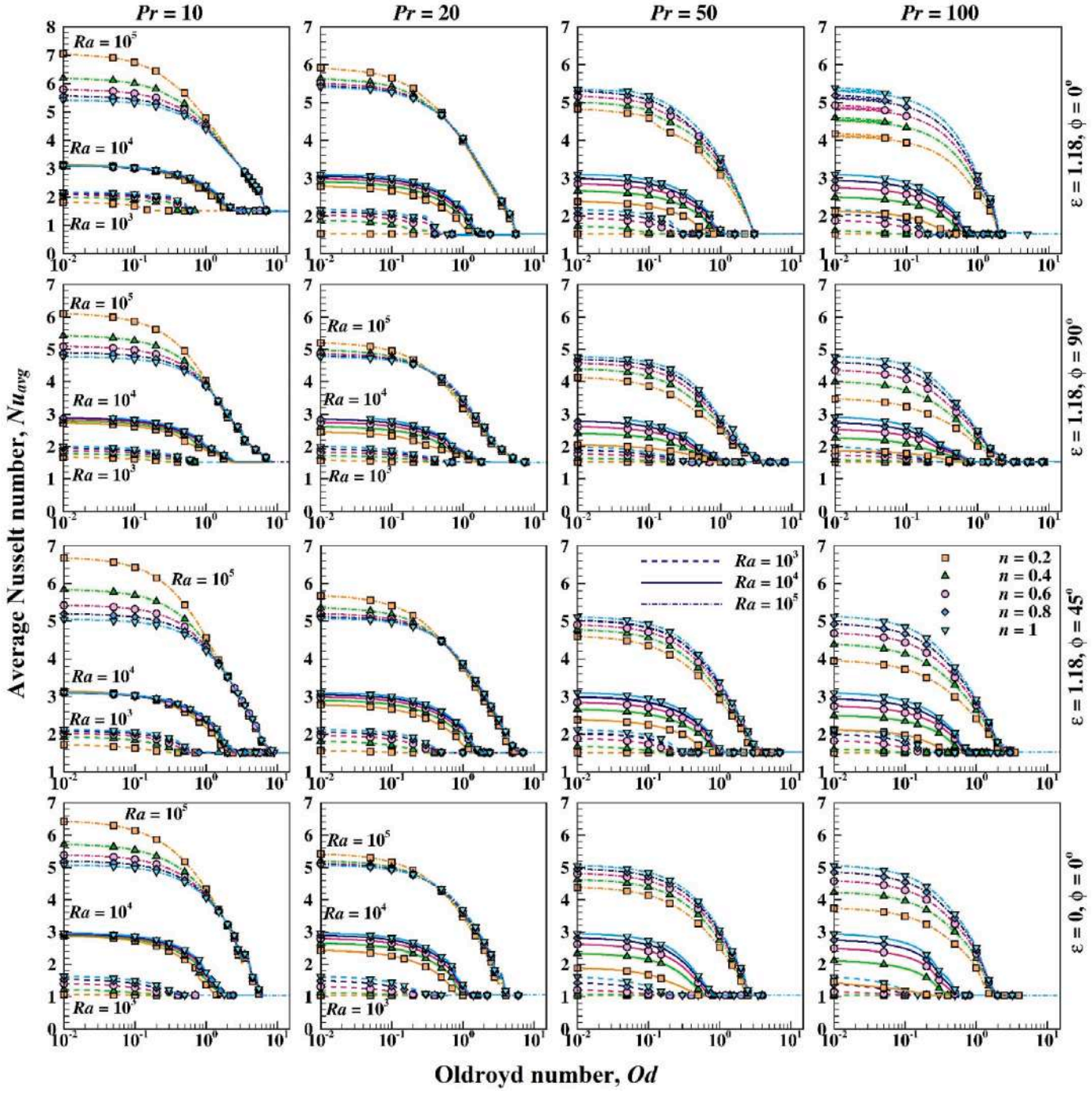


Fig. 12. The variation of average Nusselt number (Nu_{avg}) for different values of *Ra*, *Pr*, *n* and four different combinations of ε, ϕ.

local Nusselt number over the inner cylinder (heated) surface are shown in Fig. 11 at $Ra = 10^5$, $Pr = 10$, 100 and for different values of *n*, *Od*, ε and ϕ. As expected, the local Nusselt number profiles exhibit symmetry about the vertical centreline at $\phi = 0^\circ$ and asymmetric distribution can be observed at $\phi = 45^\circ$, $\phi = 90^\circ$, and $\phi = 135^\circ$ for eccentric configuration. The asymmetric effects diminish as the Oldroyd number increases and hence the symmetric profiles of local Nusselt number are observed at the limiting Oldroyd number value (Od_{max}). The symmetric nature of the local Nusselt number profiles is due to the dominance of conduction for heat transfer in the unyielded zone at Od_{max}. In line with the previous findings [33–35], the Prandtl number has visibly no influence on the convective heat transfer rate from the cylinder surface for a Newtonian fluid ($n = 1$, $Od = 0$) as its effect is included in the Rayleigh number. However, significant effects are observed for shear-thinning viscoplastic fluids. Regardless of the values of ε and ϕ, the Nusselt number decreases by ~50% as the Prandtl number increases by a factor of 10, i.e., from 10 to 100 at $n = 0.2$, $Ra = 10^5$, $Od \leq 1$. This is ascribed to the reduction in Grashof number, resulting in the weakening of the buoyancy induced currents. Similarly, the influence of Prandtl number is more prominent at high values of *Ra* compared to that at low *Ra* ($Ra = 10^3$). This finding is in line with the assertion of Mack and Bishop [12].

Due to the upward motion of the fluid on account of the buoyancy effect, the fluid below the heated cylinder is by and large stagnant and transmits heat to the fluid only by conduction. Thus, the temperature of the fluid is only slightly above $\Theta = 0$. This accounts for the large temperature gradient and hence the large values of Nu_θ are observed at the

Table 5
a: Values of the average Nusselt umber in the conduction limit. b: Value of constants in Eq. (17) and Eq. (20).

ε	Nu_∞	
0	1.0451	
0.5	1.0978	
0.7	1.1559	
1.18	1.5192	
ϕ	a	b
0°	0.125	0.252
45°	0.092	0.571
90°	−0.065	0.489

bottom surface ($\theta = 0°$ to $\theta = 180°$) of the heated cylinder. For a viscoplastic fluid in the range of Oldroyd number values $0 \leq Od \leq 0.1$, the Nu_θ is maximum at $\theta = 90°$, it decreases along the cylinder surface and shows a minimum at $\sim \theta = 270°$. This is also in agreement with the findings of Masoumi et al. [35]. This trend is generally always present irrespective of the position of the heated cylinder in the annulus. However, an increase in the Oldroyd number ($Od > 0.1$) results in the formation of unyielded zones at $\theta = 90°$ (as discussed above in section 4.2) and hence the Nusselt number decreases at $\theta = 90°$ due to the conduction-controlled heat transfer, as can be seen in the local profiles at $Od = 1$ (Fig. 11). The decrease is sharper for the case of $\varepsilon = 1.18$ and is mainly attributed to two factors, i.e., the heated cylinder is close to the outer cold cylinder and the lack of fluid motion at $\theta = 90°$ in the small intervening space between the two cylinders. Furthermore, for yield pseudoplastic fluids, the shear-dependent viscosity significantly contributes to the decrease in Nu_θ at $\theta = 90°$. Thus, the Nusselt number at the front stagnation point is lower for shear-thinning fluids than Newtonian fluids. Also, the local Nusselt number profiles for yield pseudoplastic fluids exhibit two maxima at $\theta \sim 45°$ and $\theta \sim 135°$, respectively. The viscosity is a function of the shear rate for a shear-thinning fluid and hence varies over the cylinder surface. This, in turn impinges on the local temperature gradient and hence the local Nusselt number.

For the case of a concentric annulus, the Nu_θ varies little (of the order of 10^{-4}) along the heated cylinder surface at the limiting value of Od_{max}. However, the variations increase for the eccentric placement of the heated inner cylinder; Greater is the displacement of the inner cylinder from the concentric location, i.e., higher is the value of ε, greater is the variation in the Nu_θ. The maxima are observed at $\theta = 90°$ for $\varepsilon = 1.18$, $\phi = 0°$ and $\theta = 180°$ for $\varepsilon = 1.18$, $\phi = 90°$. These are the positions closest to the outer cold cylinder, and when the entire fluid is unyielded, these positions contribute to the enhanced heat transfer rate due to the large temperature gradients at these locations.

Finally, further understanding of the relationship of the overall heat transfer rate with the non-dimensional parameters (Ra, Pr, n, Od) and the geometric parameters (ε and ϕ) can be developed by examining the results shown in Fig. 12 for the surface average Nusselt number. As anticipated, the Nu_{avg} bears a positive relationship with the Rayleigh number. The flow behaviour index influences the results via the values of Pr and/or Gr. For $Gr \leq 10^3$, the shear-thinning fluid behaviour results in reduction (~5% at $Gr = 10^3$ to ~35% at $Gr = 10$ at $Od = 0.01$) of Nu_{avg} compared to that for the case of a Newtonian fluid. For $Gr > 10^3$, enhanced heat transfer rates are observed for shear-thinning fluids. As

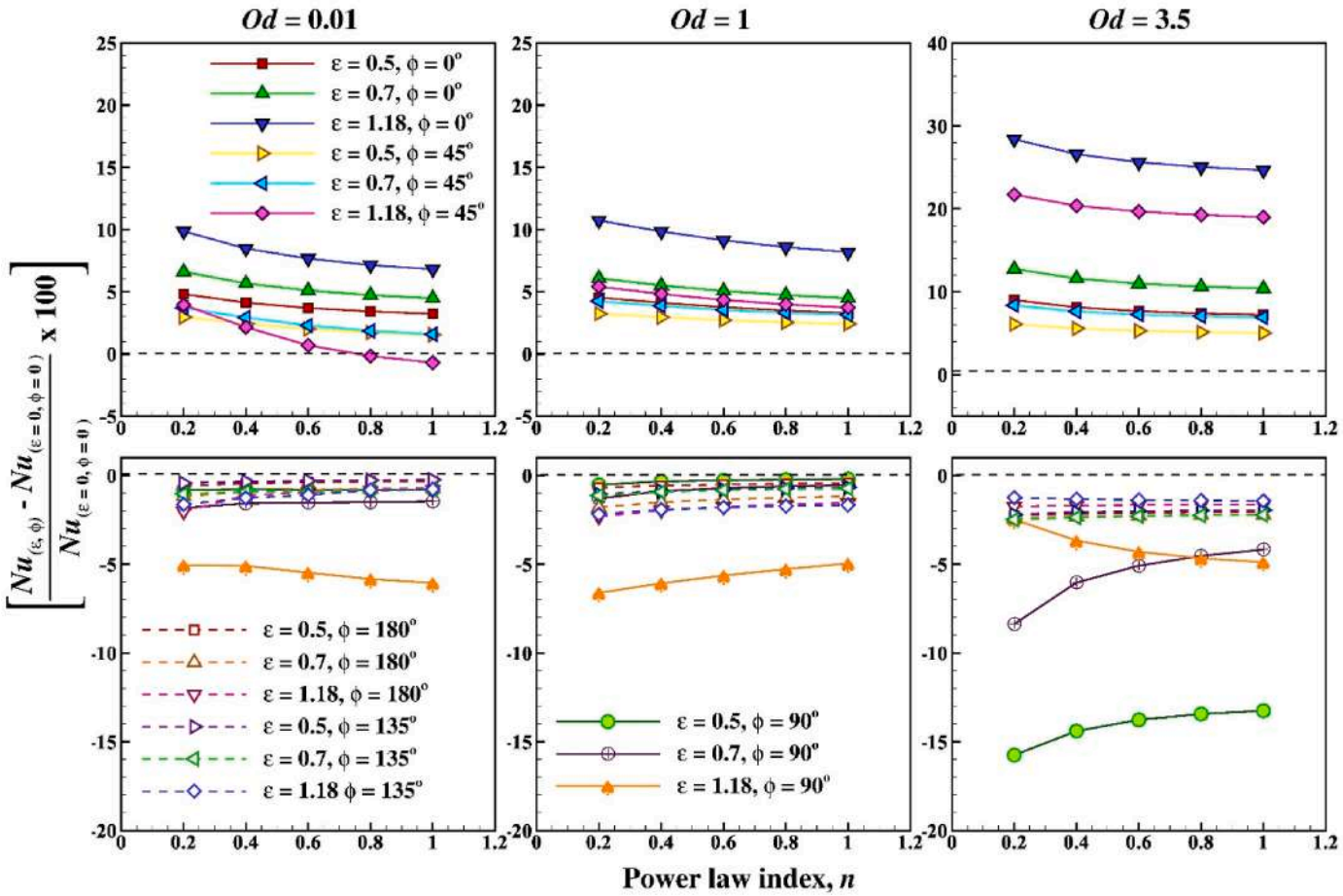


Fig. 13. Average Nusselt number for different geometric configurations (different values of ε and ϕ) scaled with that for the base case of concentric annulus ($\varepsilon = 0$, $\phi = 0°$) at $Ra = 10^5$, $Pr = 10$.

Od increases, the Nu_{avg} gradually decreases reaching the conduction limit at the fully plastic limit ($Od = Od_{max}$). The conduction limit of the average Nusselt number (Nu_{∞}) depends only on the geometric parameters, i.e., the value of ε, as summarized in Table 5a.

The effect of the geometric position of the heated cylinder is shown in Fig. 13 where the increase in the average Nusselt number relative to the corresponding value for $\varepsilon = 0$, $\phi = 0^{\circ}$ are shown at $Ra = 10^5$, $Pr = 10$. The Nusselt number is higher for $\phi = 0^{\circ}$ and $\phi = 45^{\circ}$ (all values of ε) compared to the case of $\varepsilon = 0$, $\phi = 0^{\circ}$. Correspondingly, the Nusselt number for $\phi = 90^{\circ}$, 135° and $\phi = 180^{\circ}$ is lower than that for case of concentric annulus. A clear influence of ε and ϕ can be discerned at large Rayleigh number corresponding to strong advection and the flow field is strongly influenced by the flow geometry. At $\phi = 0^{\circ}$, the Nu_{avg} increases with the downward movement of the heated cylinder (as ε increases). As discussed earlier, the vertically downward shifting provides more space for the development of stronger convective currents in the fluid above the cylinder thereby positively influencing the heat transfer rate. On the contrary, an upward shift will thereby result in a reduced rate of heat transfer as observed for the case of $\phi = 135^{\circ}$ and $\phi = 180^{\circ}$.

The horizontal shifting of the inner cylinder results in lowering the convective heat transfer rate. Thus, for $\phi = 90^{\circ}$, the surface average Nusselt number reduces as the value of ε increases. Correspondingly, minimum heat transfer is obtained at $\varepsilon = 1.18$, $\phi = 90^{\circ}$. As seen in Fig. 13, the Nusselt number reduces by ~5–7% at $Od = 0.01$ for $\varepsilon = 1.18$, $\phi = 90^{\circ}$ compared to the case of $\varepsilon = 0$, $\phi = 0^{\circ}$. This is further accentuated by an increase in the Oldroyd number (~20–30% reduction observed at $Od = 3.5$). Overall, the highest augmentation in heat transfer is obtained at $\varepsilon = 1.18$ (the inner cylinder at its lowest position below the axis), $\phi = 0^{\circ}$ when compared to the case of the concentric annulus. The heat transfer improves by 10–30% for $\varepsilon = 1.18$, $\phi = 0^{\circ}$ at $Ra = 10^5$, $Pr = 10$ depending upon the value of the Oldroyd number.

From an application standpoint, it is useful to consolidate the present values of the average Nusselt number in the form of the following correlation:

$$Nu_{avg} = Nu_{\infty} + 0.1389 Gr^{*0.1835}\, Pr^{*1.941} \frac{(Od_{max} - Od)^{0.902}}{(1 + Od)^{0.603}} (1 + \varepsilon)^{a} \tag{17}$$

where Gr^* and Pr^* are the modified Grashof number and Prandtl number [46] respectively, defined as follows,

$$Gr^* = Gr^{1/2(n+1)} \tag{18}$$

$$Pr^* = Pr^{\,n/(3n+1)} \tag{19}$$

The value of the limiting Od_{max} can be estimated by the following expression,

$$Od_{\max} = \frac{0.0443 Ra^{0.5664} n^{0.0966} (1 + \varepsilon)^{b}}{\left(1 + Pr^{0.5863}\right)} \tag{20}$$

The values of coefficient *a* and *b* appearing in Eq. (17) and Eq. (20) are tabulated in Table 5b. Eq. (17) reproduces the present numerical results (a total of 7264 data points) with the average and maximum errors of 9.77% and 38.26% respectively. Only 10% data points show errors larger than 25%. An assessment of the data revealed that the data points corresponding to low Grashof values and high Oldroyd numbers, (i.e., in the conduction limit) contributed to large errors. Eq. (20) reproduces the numerical results of Od_{max} (600 data points) with an average and maximum error of 13.7% and 48.8% respectively. Also, it is important to note here that the values of coefficients *a* and *b* capture the observed trends for the influence of geometric parameters (ε and ϕ, i.e., position of the inner cylinder) on the average Nusselt number and maximum Oldroyd number.

Table 6
Mapping of steady (S) and unsteady (U) flow regimes as a function of *Od*, *Pr*, *n*, ε, ϕ at $Ra = 10^6$.

	Od	*Pr*	*n* = 0.2	*n* = 0.4	*n* = 0.6	*n* = 0.8	*n* = 1
$\varepsilon = 0$, 0.5 (All ϕ)	0.01	10	S	S	S	S	S
		100	S	S	S	S	S
	1	10	S	S	S	S	S
		100	S	S	S	S	S
$\varepsilon = 0.7$, $\phi = 0^{\circ}$	0.01	10	U	U	U	U	U
		100	U	U	S	S	S
	1	10	U	U	S	S	S
		100	S	S	S	S	S
$\varepsilon = 1.18$, $\phi = 0^{\circ}$	0.01	10	U	U	U	U	U
		100	U	U	U	U	U
	1	10	U	U	U	U	U
		100	S	S	S	S	S
$\varepsilon = 1.18$, $\phi = 90^{\circ}$	0.01	10	U	U	U	U	U
		100	S	S	S	S	S
	1	10	S	S	S	S	S
		100	S	S	S	S	S

4.5. Loss of steady flow regime

Though the present work is concerned primarily with the steady flow regime, few transient simulations have also been performed to ascertain that the flow and thermal fields are indeed steady in the present range of conditions and to identify the conditions for the loss of steady flow regime. Naturally, now the time-dependent terms in the momentum and energy equation are retained and are solved with a non-dimensional time-step size of 0.01. The transient simulations were initiated by using the steady state solutions based on the values of the parameters to avoid potential convergence difficulties. Based on a detailed examination of the results obtained with the steady and transient simulations, the flow remains steady irrespective of the values of *Od*, *Pr*, *n*, ϕ and ε (eccentricity) up to $Ra = 10^5$. However, the flow loses its steady characteristics beyond $Ra = 10^5$ depending on the values of the other parameters, as shown in Table 6 (as *S* or *U* implying steady or unsteady flow regime). The flow remains steady for the case of concentric ($\varepsilon = 0$) and the eccentric ($\varepsilon = 0.5$) cases even at $Ra = 10^6$, regardless of the values of *Od, n, Pr* and ϕ. Thereafter, the transient behaviour sets in for the eccentricity values greater than 0.5 ($\varepsilon > 0.5$). Similarly, an increase in the value of ϕ also postpones the onset of the unsteady regime. In contrast, the shear-thinning fluid behaviour ($n < 1$) accelerates the transition to the unsteady flow regime. The Oldroyd number stabilizes the flow and deters the onset of the transient flow characteristics. Thus, the flow remains steady over the entire envelope of conditions examined here for $Od \geq 2$ at $Ra = 10^6$.

5. Conclusions

In this work, numerical results on free convection in a circular annulus filled with a yield-pseudoplastic fluid have been obtained in the two-dimensional, steady, laminar flow regime over the range of kinematic and geometric parameters as: $10^3 \leq Ra \leq 10^6$, $10 \leq Pr \leq 100$, $0.2 \leq n \leq 1$, $0 \leq Od \leq Od_{max}$, $0 \leq \varepsilon \leq 1.18$ and $0^{\circ} \leq \phi \leq 180^{\circ}$. The individual contribution of each parameter to the convective heat transfer has been studied in detail. The shear-thinning fluid viscosity improves the convective heat transport only at values of $Gr > 10^3$ when there is a fair degree of advection present. The limiting value of Oldroyd number to achieve the fully plastic limit is higher for the eccentric annuli than that for a concentric annulus. This affords an opportunity of fluid yielding over broader ranges of kinematic parameters than that for a concentric annulus. Overall, the maximum possible augmentation in the average Nusselt number of ~30% is observed at $\varepsilon = 1.18$, $\phi = 0^{\circ}$ at $Ra = 10^5$, $Pr = 10$ compared to the corresponding case of concentric annulus. On the contrary, the horizontal shifting of the heated cylinder results in the lowering of heat transfer due to the constricted fluid flow. Thus, the

surface average Nusselt number reduces by up to 5–20% at $\varepsilon = 1.18$, $\phi = 90^\circ$ depending upon the values of *Ra*, *Pr*, *n* and *Od* compared to the case of a concentric annulus. Limited results of the critical Rayleigh number for the cessation of the steady flow regime are also presented as a function of the pertinent dimensionless groups. The present values of the average Nusselt number have been consolidated in the form of a correlation to facilitate its use in future applications.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

No data was used for the research described in the article.

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